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# Pooling and Drift in Delayed Bandits

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## Abstract

A learner chooses an action, observes an intermediate state immediately, and receives a bounded outcome only after a fixed delay. When the outcome mean depends on the action only through that state, one delayed observation can inform every action that could have produced it. We quantify this sharing by an *effective dimension*  $v_t \in [1, |\mathcal{S}|]$ , set by the overlap of the action-induced state distributions under the learner’s current play; its worst case  $\bar{v} = \sup_x v(P, x)$  depends on  $P$  alone. For known  $P$ , the analyzed  $m = d + 1$  version of STATE-EXP3 attains  $\tilde{O}(\sqrt{(d+1)V \log K})$  for any deterministic  $V \geq \sum_t v_t$  fixed before the run. Coarsening the states lowers the dimension at the price of an explicit within-group bias. A separate temporal cost remains: even a learner given the exact stale action-loss vector  $m_t = c_{t-d}$  suffers  $\Omega(\sqrt{dE_2 \min\{1 + \log q, T/d\}})$  on an appropriate oblivious instance, where  $E_2 = \sum_t \|c_t - m_t\|_\infty^2$ . Experiments on the unproved but practical  $m = 1$  variant motivate a conjectured sharper delay dependence, showing larger gains as overlap increases at matched difficulty and a regret reduction of as much as 79 per cent in a hand-structured synthetic funnel. The results separate cross-action pooling from the independently persistent cost of temporal staleness.

## 1 Introduction

Delayed outcomes are common in sequential decision-making. A system must choose an action now even though the outcome it cares about may arrive much later. Recommendation is a representative example. An item is shown immediately, an engagement signal such as a click or browsing depth is observed soon after, and conversion or retention may only be known days or weeks later. The central question here is when such an intermediate signal can reduce the cost of learning before the final outcome arrives.

In this interpretation, actions are catalogue items, states are intermediate engagement outcomes, and the delayed outcome is conversion. Different items may induce similar distributions over engagement states, so an outcome observed after one item can carry information about many others. The difficulty scales not with the catalogue size  $K$  but with how differently the actions distribute users across the intermediate states. Operationally, a catalogue with thousands of items can still be easy to learn from if those items funnel users through only a few genuinely different engagement patterns.

Bandits with intermediate observations formalize this. Esposito et al. (2023) show the state-to-loss map decides the complexity, with an unrestricted one giving only the rate available with no intermediate observations.

Because similar actions reach similar states, we can *pool through the state*, charging a delayed outcome to every action that could have produced the state observed. The resulting statistical cost is governed by an *effective dimension*  $v_t$ , which measures how distinguishable the action-induced state distributions are under the learner’s current play. When many actions behave similarly through the state channel,  $v_t$  can be far smaller than  $K$ .

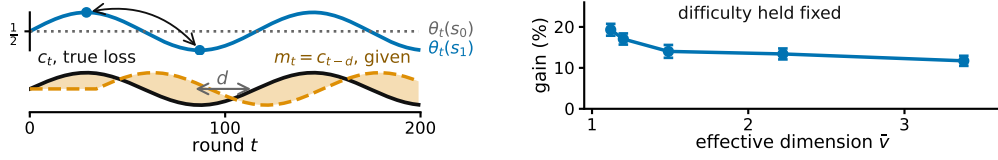


Figure 1: **Both panels synthetic.** **Left, the state-to-outcome relationship changes while the action-to-state channel stays fixed**, so the loss vector from  $d$  rounds earlier grows increasingly inaccurate. This discrepancy is measured by  $E_2$ . **Right, more overlap allows more sharing across actions**, at matched task difficulty,  $K = 40$ ,  $|\mathcal{S}| = 8$ ,  $T = 10000$ ,  $d = 50$ , 20 paired seeds, bars one standard error of the paired difference. **The right panel runs the practical  $m = 1$  variant.** **Theorem 1 analyzes  $m = d + 1$  and does not prove this, so the panel motivates Conjecture 4** (Appendix I, study 10).

Delay creates a second difficulty even if the learner is handed an exact description of the past. Let  $m_t = c_{t-d}$  be the action-loss vector from  $d$  rounds earlier. Its usefulness depends on how far the losses have moved, which we measure by  $E_2 = \sum_t \|c_t - m_t\|_\infty^2$  (Figure 1). Our lower bound asks how much regret this mismatch forces even when  $m_t$  is supplied for free. Vernade et al. (2020) vary the action-to-state map while fixing the state-to-loss map, whereas we fix the former and let the latter vary obliviously.

Intermediate feedback helps *across actions*, where shared states reduce the independent exploration needed, but cannot remove the cost incurred *across time*, where even perfectly known past information can go stale before the next decision.

| Question                             | Setting                | Answer                                                    |
|--------------------------------------|------------------------|-----------------------------------------------------------|
| Can states cut the cost of learning? | $P$ known, $m = d + 1$ | Thm 1: $\tilde{O}(\sqrt{(d+1)V \log K})$                  |
| Can we simplify the representation?  | $g$ fixed in advance   | Thm 2: $V(g)$ against $2 \sum_t \delta_t(g)$              |
| Can it remove the cost of delay?     | exact $m_t$ supplied   | Thm 3: no, $\Omega(\sqrt{d E_2 \min\{1 + \log q, T/d\}})$ |

Together, the three results separate two difficulties that intermediate feedback creates. Theorems 1 and 2 isolate how overlap in a fixed action-to-state map reduces the cost of learning across actions, whereas Theorem 3 shows that stale information can remain costly across time even when an exact delayed loss vector is supplied for free. Unknown  $P$  is estimable with a conservative warm-start guarantee (Appendix E), while jointly adapting to overlap and temporal drift remains open (Conjecture 4).

## 2 Related work

*Delayed bandits.* Zimmert and Seldin (2020) give the optimal  $\tilde{O}(\sqrt{KT} + \sqrt{D \log K})$  for total delay  $D$ , and Esposito et al. (2023) allow an unknown and possibly adversarial action-to-state map whereas ours is fixed and, for the upper bounds, known. *Stale predictions.* Bar-On and Mansour (2025) derive upper bounds adapting to the inaccuracy of evolving observations. Our lower bound instead asks what delay-dependent cost is unavoidable as a function of stale-prediction error, while our upper bound separately exploits the geometry of a fixed action-to-state map. Wang et al. (2026) bound squared gradient prediction error without delay, so the delay factor and its saturation are new here. Zhang et al. (2026) share the motivation in a stochastic Bayesian model whereas ours is adversarial. *Mediator feedback* provides the closest structural connection:  $v_t - 1$  is the  $\chi^2$  mutual information appearing in that literature (Eldowa et al., 2024). The capacity quantity is known from mediator feedback; our contribution is its role when the outcome attached to the observed state is delayed, and the separation of that structural benefit from an independent temporal obstruction (Appendix F). Feedback graphs and off-policy coverage (Gabbianelli et al., 2023) provide related forms of information sharing; Appendix A gives the detailed comparison.

### 3 Problem formulation

There are  $T$  rounds,  $K$  actions, a finite state space  $\mathcal{S}$ , and a fixed known delay  $d$ ; all logarithms are natural. Let  $\mathcal{F}_{t-1}$  denote all information available immediately before the learner chooses  $A_t$ . At round  $t$  the learner draws  $A_t$  from an  $\mathcal{F}_{t-1}$ -measurable  $x_t$ , observes  $S_t \sim P(\cdot | A_t)$  at once, and at time  $t + d$  observes  $X_t \in [0, 1]$  with  $\mathbb{E}[X_t | \mathcal{F}_{t-1}, A_t, S_t] = \theta_t(S_t)$ , outcomes conditionally independent across rounds given the states. The outcome  $X_r$  arrives after the round- $r$  decision and may first affect the learner's action distribution at round  $r + d + 1$ ; equivalently, the round- $t$  decision uses realized outcomes only from rounds  $r \leq t - d - 1$ . The conditional mean depends on  $A_t$  only through  $S_t$ . This *state-sufficiency* assumption is exactly what licenses cross-action pooling: after observing an outcome at state  $S_t$ , the learner may use it to update every action that could have produced that state. If an action has a direct effect on the outcome not represented in  $S_t$ , the argument no longer applies.

The *action-to-state matrix*  $P \in [0, 1]^{K \times |\mathcal{S}|}$  is row-stochastic and fixed, and  $\theta_t \in [0, 1]^{|\mathcal{S}|}$  is an oblivious sequence of *state-loss means*, so the induced action losses are  $c_t = P\theta_t$ . Performance is the static pseudo-regret  $\mathbb{E}[R_T] = \mathbb{E}[\sum_t c_t(A_t)] - \min_a \sum_t c_t(a)$ . For the lower bound only, the learner also receives the *stale action-loss vector*  $m_t \triangleq c_{t-d}$  before choosing  $A_t$ , with  $m_t = c_1$  for  $t \leq d$ . This is not a function of the observables, so a lower bound surviving it is stronger, and STATE-EXP3 never uses it. We measure how misleading  $m_t$  is by  $E_2 = \sum_t \|c_t - m_t\|_\infty^2 \leq T$ , which stays small when errors are small though frequent.

### 4 Pooling through the state

Here  $P$  is known and the learner does not receive  $m_t$ . Write  $q_t(s) = \sum_a x_t(a)P(s | a)$  and set  $\hat{c}_t(a) = P(S_t | a)X_t/q_t(S_t)$ , which is unbiased for  $c_t(a)$  and whose play-weighted second moment is at most the *effective dimension*  $v_t = \sum_s q_t(s)^{-1} \sum_a x_t(a)P(s | a)^2 \in [1, |\mathcal{S}|]$ , equal to one plus the  $\chi^2$  mutual information between action and state. STATE-EXP3 runs  $m$  interleaved copies of exponential weights / EXP3 (Auer et al., 2002), copy  $i$  taking rounds congruent to  $i$  modulo  $m$ , playing  $x_t \propto \exp(-\eta L_i)$  and adding  $\hat{c}_t$  to  $L_i$  once  $X_t$  is usable. Each update costs  $O(K)$  (Appendix C, with pseudocode).

**Theorem 1.** *Let  $P$  be known and  $(\theta_t)$  oblivious. For every  $\eta > 0$ , STATE-EXP3 with  $m = d + 1$  satisfies  $\mathbb{E}[R_T] \leq (d + 1) \log K / \eta + \frac{\eta}{2} \mathbb{E}[\sum_t v_t]$ , so any deterministic  $V$  fixed before the run with  $\sum_t v_t \leq V$  surely and  $\eta = \sqrt{2(d + 1) \log K / V}$  give  $\mathbb{E}[R_T] \leq \sqrt{2(d + 1)V \log K}$ . Writing  $\bar{v}$  for the largest  $v_t$  over all play distributions,  $V = \bar{v}T \leq |\mathcal{S}|T$  is such a bound and depends on  $P$  alone.*

The effective dimension measures how strongly the current play distinguishes actions through the state channel:  $v_t$  approaches one when the action-induced state distributions overlap heavily, and grows when the observed state reveals which action was played. The proof combines two facts: the pooled estimator is unbiased with play-weighted second moment at most  $v_t$ , and the  $d + 1$  interleaved copies make each copy's own feedback immediate. Formally, with  $V = \bar{v}T$ , Theorem 1 improves on the unrestricted-state-to-loss rate  $\tilde{O}(\sqrt{(K + d)T})$  (Esposito et al., 2023) when  $\bar{v}(d + 1) \log K \lesssim K + d$ . Theorem 1 isolates the benefit of cross-action sharing but is neither delay-adaptive nor tight at every structural endpoint. Interleaving makes the delay cost multiplicative in  $d + 1$ , the bound pays  $v_t$  rather than  $v_t - 1$  and so does not vanish when all rows of  $P$  agree, and it does not adapt to  $E_2$ . These gaps motivate the  $m = 1$  variant and Conjecture 4. A best-state rule can also be stronger under an additional structural assumption, but suffers linear regret when that assumption fails (Appendix I).

### 5 The representation trade-off

Fix  $g : \mathcal{S} \rightarrow \mathcal{Z}$  and run the same algorithm on  $g(S_t)$  with  $P^g(z | a) = \sum_{s: g(s)=z} P(s | a)$ . Coarsening never raises the dimension, but the grouped estimator tracks the group average of  $\theta_t$  and acquires a bias  $\delta_t(g) = \max_z \max_{s, s': g(s)=g(s')=z} |\theta_t(s) - \theta_t(s')|$ .

**Theorem 2.** *Under the hypotheses of Theorem 1, running on  $g$  with  $\sum_t v_t(g) \leq V(g)$  gives  $\mathbb{E}[R_T] \leq \sqrt{2(d + 1)V(g) \log K} + 2 \sum_t \delta_t(g)$ .*

Theorem 2 makes the state representation a bias–variance design choice. Coarsening need not preserve the original state-sufficiency condition: the grouped estimator is instead unbiased for a predictable surrogate loss whose per-action discrepancy from  $c_t$  is at most  $\delta_t(g)$  (Appendix D). Thus finer representations preserve more decision-relevant distinctions but raise statistical complexity, whereas coarser representations share more aggressively and pay the explicit  $2 \sum_t \delta_t(g)$  bias term. The map  $g$  is fixed in advance; learning it from data remains open.

## 6 The cost of drift

We now ask how much drift costs even when the stale action losses are free. Theorem 3 isolates a temporal obstruction, not a joint overlap–drift bound: the construction gives the  $q$  drifting actions private states, removing useful pooling and forcing  $q \leq |S| - 1$  (Appendix G).

**Theorem 3.** *Fix  $T$ ,  $1 \leq d \leq T/2$ ,  $\mathcal{E} \leq T/4$ , and  $1 \leq q \leq \min\{K - 1, |S| - 1\}$ . For every possibly randomized algorithm, including one given the exact stale action-loss vector, there exists an oblivious instance with  $E_2 \leq \mathcal{E}$  such that  $\mathbb{E}[R_T] = \Omega(\sqrt{d\mathcal{E} \min\{1 + \log q, T/d\}})$ . If the oracle is withheld and  $K \leq T$ , the same model class has minimax regret  $\Omega(\sqrt{KT} + \sqrt{d\mathcal{E} \min\{1 + \log q, T/d\}})$ , up to absolute constant factors.*

A decision-maker told the exact loss vector from  $d$  rounds ago can still choose systematically badly when the environment moves fast enough.

In the oracle-free lower bound,  $\sqrt{KT}$  is the ordinary exploration cost, whereas the second term depends on the delay  $d$ , the stale-prediction error  $E_2$ , and the number  $q$  of independently drifting coordinates available to the comparator. The two terms are established using different hard instances; taking their maximum yields the displayed additive form up to an absolute factor of two.

## 7 What the experiments show

The main experiments study the practical  $m = 1$  variant. They are designed to test the pooling mechanism and the qualitative role of overlap, not to validate Theorem 1, whose proof requires  $m = d + 1$ . We therefore read them as evidence motivating Conjecture 4; Appendix I evaluates the analyzed variant.

A synthetic recommendation funnel illustrates the regime. With  $K = 200$  catalogue items and six engagement states the estimated effective dimension is only 1.97, and the state-pooled variant cuts regret against action-level weighting by 79.3, 63.6 and 38.9 per cent at  $d = 10, 50, 200$ . The funnel is hand-structured rather than fitted to data, so the results illustrate the mechanism, not a deployed system. On a Dirichlet family it gains 5 to 22 per cent as  $K$  grows from 8 to 80, and an online plug-in  $\hat{P}_t$  stays within 0.5 per cent of known  $P$ .

Two negative results are reported in full. A best-state heuristic leads seven of nine settings, though its regret grows linearly where its assumption fails, and certified Safe-Pool pools nothing in any run (Appendix I).

## 8 Discussion

The results suggest a simple diagnostic. Pooling helps most when many actions induce similar state distributions, so the effective dimension falls well below the action count. Coarsening strengthens it when grouped states have similar outcome means, while fast-changing state-to-outcome relationships limit how far delayed information can be trusted. The question is not whether an intermediate signal exists, but whether it is shared across actions and stays informative over the delay.

**Conjecture 4** (open). *STATE-EXP3 with  $m = 1$  satisfies the first displayed bound, and some algorithm achieves the second. Both statements remain unproved.*

$$\mathbb{E}[R_T] = \tilde{O}(\sqrt{V \log K} + \sqrt{dT \log K}), \quad \mathbb{E}[R_T] = \tilde{O}(\sqrt{KT} + \sqrt{d E_2 \log K}).$$

STATE-EXP3 formalizes the first requirement; Theorem 3 shows the second can impose an unavoidable cost. The results separate the *cross-action* and *cross-time* difficulties but do not yet yield a single algorithm adapting to both, the main open problem left here (Appendices H, J).

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## 205 A Extended related work

206 The minimax characterization of delayed adversarial bandits traces to Cesa-Bianchi et al. (2019),  
 207 and Joulani et al. (2013) give the additive form of the delay penalty in the stochastic case. On acting  
 208 before the loss is known, Flaspohler et al. (2021) study optimism under delay directly, Wei et al.  
 209 (2020) delimit when prediction error controls regret, Hsieh et al. (2022) show that an optimistic step  
 210 must run at a larger scale than the update, and Zhang (2026) gives second-order path-length bounds.

211 Structure that lets one observation inform several actions is the other half of the picture. Feedback  
 212 graphs (Alon et al., 2015) fix which actions are revealed together, and their stochastic version (Es-  
 213 posito et al., 2022) randomizes that set. Factored rewards with intermediate observations (Mussi  
 214 et al., 2024) decompose the reward instead. Off-policy coverage (Gabbianelli et al., 2023) is the  
 215 closest analogue of our effective dimension, since its mismatch coefficient measures the same kind  
 216 of overlap between the distribution that generated the data and the one being evaluated. The dis-  
 217 tinction from all of these is that a feedback graph randomizes *who* is observed, whereas our sensor  
 218 randomizes *what* is observed and the sharing is a consequence of  $P$  rather than a modeling primitive.

## 219 B Notation

Table 1: Symbols used in the paper. Every symbol is also defined at first use in the body, so this table is a reference rather than a dependency.

| Symbol                                  | Meaning                                                                                              |
|-----------------------------------------|------------------------------------------------------------------------------------------------------|
| $T, K, d$                               | rounds, actions, fixed feedback delay                                                                |
| $\mathcal{S}, S_t$                      | finite state space, state observed at round $t$                                                      |
| $P \in [0, 1]^{K \times  \mathcal{S} }$ | row-stochastic action-to-state matrix, fixed known in Theorems 1 and 2, estimated in Appendix E      |
| $\theta_t \in [0, 1]^{ \mathcal{S} }$   | state-loss means at round $t$ , oblivious                                                            |
| $c_t = P\theta_t \in [0, 1]^K$          | induced loss over actions                                                                            |
| $q_t(s) = \sum_a x_t(a)P(s   a)$        | state distribution induced by the play                                                               |
| $X_t \in [0, 1]$                        | outcome at time $t + d$ , $\mathbb{E}[X_t   \mathcal{F}_{t-1}, A_t, S_t] = \theta_t(S_t)$            |
| $x_t, A_t$                              | action distribution at round $t$ , action drawn from it                                              |
| $m_t = c_{t-d}$                         | stale action-loss vector, given before the action, $m_t = c_1$ for $t \leq d$                        |
| $E_2$                                   | $\sum_t \ c_t - m_t\ _\infty^2$ , accumulated squared prediction error                               |
| $\Lambda_2$                             | $\sum_t \min\{1, \ c_t - m_t\ _2^2\}$ , the measure of Bar-On and Mansour (2025)                     |
| $W$                                     | $\sum_t \ \theta_t - \theta_{t-1}\ _\infty^2$ , state-loss variation                                 |
| $\hat{c}_t, \tilde{c}_t, \check{c}_t$   | state-pooled, action-level, centered loss estimates                                                  |
| $\hat{g}_t$                             | the Safe-Pool blend $(1 - \lambda_t)\tilde{c}_t + \lambda_t\hat{c}_t$ , a distinct object            |
| $\bar{c}_t, c_t^g, b_t$                 | conditional mean of $\hat{c}_t$ , grouped surrogate loss, $\bar{c}_t - c_t$                          |
| $v_t, \bar{v}, V$                       | effective dimension $\sum_s q_t(s)^{-1} \sum_a x_t(a)P(s   a)^2$ , its supremum, its budget          |
| $\hat{v}$                               | Monte Carlo estimate of $\bar{v}$ , used for measured values in Appendix I                           |
| $g, \mathcal{Z}, \delta_t(g)$           | state grouping, its range, widest within-group loss spread                                           |
| $\mathcal{E}, q$                        | budget constraining $E_2$ , number of independently drifting coordinates                             |
| $h, B, \varepsilon, \sigma_b^{(j)}$     | block length, block count, amplitude, Rademacher signs                                               |
| $\eta, \beta, C$                        | learning rate, Tsallis parameter, residual scale                                                     |
| $\alpha$                                | smoothing pseudocount in $\hat{P}_t$                                                                 |
| $\kappa$                                | sensor balance, $\bar{p}(s) \geq 1/(\kappa \mathcal{S} )$ for $\bar{p} = K^{-1} \sum_a P(\cdot   a)$ |
| $m, L_i, G_r$                           | interleaving factor, estimate of copy $i$ , estimates outstanding at $r$                             |
| $D, d_{\max}, \bar{\Delta}, \Gamma$     | total delay, maximal delay, inaccuracy bound, the measure of Wang et al. (2026)                      |

## 220 C The state-pooled estimator

221 **The algorithm.** Inputs are the action-to-state matrix  $P$ , the delay  $d$ , the number of copies  $m$ , and  
 222 the rate  $\eta > 0$ . The analyzed setting is  $m = d + 1$  and the practical variant is  $m = 1$ . Round  $r$ 's  
 223 outcome becomes usable at round  $r + d + 1$ , which for  $m = d + 1$  is the next round belonging to  
 224 the copy that played round  $r$ .

- 225 1. Set  $L_i \leftarrow (0, \dots, 0) \in \mathbb{R}^K$  for  $i = 0, \dots, m - 1$ , and let  $\mathcal{P}$  be an empty table of rounds  
 226 awaiting their outcome.
- 227 2. For  $t = 1, \dots, T$ :

- 228 (a) *Deliver*. If  $r \triangleq t - d - 1 \geq 1$ , read  $(i_r, q_r(S_r), S_r, X_r)$  from  $\mathcal{P}$ , remove it, and update  
 229 only that copy,  $L_{i_r}(a) \leftarrow L_{i_r}(a) + P(S_r | a)X_r/q_r(S_r)$  for every  $a$ . This is the  
 230 pooled step, and it touches every action rather than the one played.  
 231 (b) *Select the copy*.  $i \leftarrow t \bmod m$ .  
 232 (c) *Play*.  $x_t(a) \propto \exp(-\eta L_i(a))$ , draw  $A_t \sim x_t$ , and observe  $S_t$  at once.  
 233 (d) *Record*. Compute the scalar  $q_t(S_t) = \sum_a x_t(a)P(S_t | a)$  and store  $(i, q_t(S_t), S_t, \cdot)$   
 234 in  $\mathcal{P}$ , to be completed when  $X_t$  arrives at time  $t + d$ .

235 Steps (a), (c) and (d) are each  $O(K)$  and there is no optimization subproblem, so a round costs  $O(K)$   
 236 once  $P$  is stored. The state is held in the  $m$  vectors  $L_i$ , which is  $mK$  numbers, the matrix  $P$ , which  
 237 is  $K|\mathcal{S}|$  numbers, and four scalars for each of the at most  $d$  rounds in flight. Only the scalar  $q_r(S_r)$   
 238 is needed from round  $r$ , not the vector  $x_r$ .

239 *Remark 5* (The  $\chi^2$  form). Under  $x_t$  the pair  $(A_t, S_t)$  has law  $x_t(a)P(s | a)$  with marginals  $x_t$  and  
 240  $q_t$ , so

$$\begin{aligned} \chi^2(P_{A_t, S_t} \| P_{A_t} \otimes P_{S_t}) &= \sum_{a, s} \frac{(x_t(a)P(s | a) - x_t(a)q_t(s))^2}{x_t(a)q_t(s)} \\ &= \sum_{a, s} \frac{x_t(a)P(s | a)^2}{q_t(s)} - 2 + 1 = v_t - 1, \end{aligned}$$

241 the cross term summing to  $\sum_{a, s} x_t(a)P(s | a) = 1$  and the last to  $\sum_{a, s} x_t(a)q_t(s) = 1$ . So  $v_t = 1$   
 242 exactly when  $A_t$  and  $S_t$  are independent, which pins the rows of  $P$  only on the support of  $x_t$  and  
 243 therefore means all rows agree whenever  $x_t$  has full support, as exponential weights does. If  $P$   
 244 is deterministic and every state is reached by some action in the support of  $x_t$ , then  $v_t$  equals the  
 245 number of states that support reaches. In particular, under full-support play and an onto deterministic  
 246  $P$ ,  $v_t = |\mathcal{S}|$ . This is an identity, not a bound, and it is used here only for reading  $v_t$ , never in a proof.

247 **Proposition 6.** Fix  $t$ , let  $x_t$  be the play distribution,  $q_t(s) = \sum_a x_t(a)P(s | a)$  the induced state  
 248 distribution, and  $\hat{c}_t(a) = P(S_t | a)X_t/q_t(S_t)$ , which is well defined because only states with  
 249  $q_t(s) > 0$  occur. For statements involving  $1/x_t(a)$ , restrict to actions with  $x_t(a) > 0$ ; STATE-EXP3  
 250 itself has full support on every action. Then  $\hat{c}_t(a) \geq 0$ ,  $\hat{c}_t(a) \leq 1/x_t(a)$ ,  $\mathbb{E}[\hat{c}_t(a) | \mathcal{F}_{t-1}] = c_t(a)$ ,  
 251 and

$$\sum_a x_t(a)\mathbb{E}[\hat{c}_t(a)^2 | \mathcal{F}_{t-1}] \leq v_t := \sum_{s: q_t(s) > 0} \frac{\sum_a x_t(a)P(s | a)^2}{q_t(s)}, \quad 1 \leq v_t \leq |\mathcal{S}|,$$

252 states of zero probability being omitted throughout, which is the convention  $0/0 = 0$ . The action-  
 253 level  $\hat{c}_t(a) = \mathbf{1}\{A_t = a\}X_t/x_t(a)$  has the same first three properties and weighted second moment  
 254 at most  $K$ .

255 *Proof.* Nonnegativity is immediate from  $X_t \geq 0$ . Given  $\mathcal{F}_{t-1}$  the state  $S_t$  has distribution  $q_t$ , and  
 256  $\mathbb{E}[X_t | \mathcal{F}_{t-1}, S_t = s] = \theta_t(s)$  by the model, since that conditional mean does not depend on  
 257 the action, so  $\mathbb{E}[\hat{c}_t(a) | \mathcal{F}_{t-1}] = \sum_s q_t(s)P(s | a)\theta_t(s)/q_t(s) = c_t(a)$ . For the range,  $q_t(s) \geq$   
 258  $x_t(a)P(s | a)$  for every  $a$ , so  $P(s | a)/q_t(s) \leq 1/x_t(a)$  and  $X_t \leq 1$ . For the second moment,  
 259  $X_t^2 \leq 1$  gives  $\mathbb{E}[\hat{c}_t(a)^2 | \mathcal{F}_{t-1}] \leq \sum_s P(s | a)^2/q_t(s)$ , and exchanging the sums gives  $v_t$ . Each  
 260 summand of  $v_t$  is at most 1, since  $P(s | a) \leq 1$  makes  $\sum_a x_t(a)P(s | a)^2 \leq q_t(s)$ , and at least  
 261  $q_t(s)$ , since Jensen applied to  $a \mapsto P(s | a)$  under  $x_t$  gives  $\sum_a x_t(a)P(s | a)^2 \geq q_t(s)^2$ . Summing  
 262 the two bounds gives  $1 \leq v_t \leq |\mathcal{S}|$ . The action-level claims are the same computation with the state  
 263 replaced by the played action, whose weighted second moment is  $\sum_a x_t(a)/x_t(a) = K$ .  $\square$

264 The two ends are attained. If every action induces the same state distribution then  $v_t = \sum_s q_t(s) =$   
 265 1, and if  $P$  is deterministic, each action reaching a single state, then every  $P(s | a)$  is 0 or 1 and  
 266 each summand is exactly 1, so  $v_t$  counts the states reached and equals  $|\mathcal{S}|$  when every state has an  
 267 action reaching it. Disjoint row supports give  $v_t = K$  rather than  $|\mathcal{S}|$ , since the summands over  
 268 one action's private states add to  $\sum_s P(s | a) = 1$ , so a sensor with many private states per action  
 269 resolves few of them. So  $v_t$  reads the overlap between the rows of  $P$  rather than their length, and  
 270  $\bar{v} = \sup_x v(x)$ , the supremum over play distributions, depends on  $P$  alone. An outcome observed  
 271 at a state updates the estimate for every action that can reach it, which is why the construction of  
 272 Section 6 gives each drifting coordinate a private state, forcing  $v_t = |\mathcal{S}|$  there. The immediately  
 273 observed pairs  $(A_t, S_t)$  identify  $P$  separately, though Theorem 1 assumes it known.

## 274 D Proofs of Theorems 1 and 2

275 **Proof idea.** The pooled estimator is unbiased for every action, while Proposition 6 bounds its play-  
 276 weighted second moment by  $v_t$  rather than  $K$ . With  $m = d + 1$  interleaved copies, the outcome  
 277 generated on a round belonging to one copy is available before that copy acts again, so each copy  
 278 is an ordinary exponential-weights process with immediate feedback. Summing the  $d + 1$  potential  
 279 inequalities gives  $(d + 1) \log K/\eta$ , and the pooled second moments give  $\frac{\eta}{2} \sum_t v_t$ .

280 *Proof of Theorem 1.* Fix a copy  $i$  and let  $T_i$  be the number of rounds it owns, so  $\sum_i T_i = T$ . Round  
 281  $t$ 's outcome is usable from round  $t + d + 1$ , which with  $m = d + 1$  is exactly  $t + m$ , the copy's next  
 282 round, so within a copy the feedback is undelayed and  $L_i$  carries exactly the estimates of that copy's  
 283 earlier rounds. Exponential weights on nonnegative losses give, for any action  $a^*$  and any  $\eta > 0$ ,

$$\sum_{t \in i} (\langle x_t, \hat{c}_t \rangle - \hat{c}_t(a^*)) \leq \frac{\log K}{\eta} + \frac{\eta}{2} \sum_{t \in i} \sum_a x_t(a) \hat{c}_t(a)^2,$$

284 by the usual potential telescoping with  $e^{-z} \leq 1 - z + z^2/2$ , which needs  $z \geq 0$  and therefore  
 285  $\hat{c}_t \geq 0$ , supplied by Proposition 6. Each  $x_t$  is  $\mathcal{F}_{t-1}$ -measurable, because  $L_i$  then holds only rounds  
 286  $r \leq t - m = t - d - 1$ , so taking expectations turns the left side into  $\sum_{t \in i} (\langle x_t, c_t \rangle - c_t(a^*))$  by  
 287 unbiasedness and the quadratic term into at most  $\mathbb{E}[\sum_{t \in i} v_t]$ , again by Proposition 6. Summing over  
 288 the  $m = d + 1$  copies and using  $\sum_i \min_a \sum_{t \in i} c_t(a) \leq \min_a \sum_t c_t(a)$ , which is what lets the copies  
 289 be compared against one common action, gives  $\mathbb{E}[R_T] \leq (d + 1) \log K/\eta + \frac{\eta}{2} \mathbb{E}[\sum_t v_t]$ . If  $\sum_t v_t \leq$   
 290  $V$  surely then  $\eta = \sqrt{2(d + 1) \log K/V}$  balances the two terms and gives  $\sqrt{2(d + 1)V \log K}$ , and  
 291  $V = \bar{v}T$  is admissible by the definition of  $\bar{v}$ .  $\square$

292 The delay enters as a factor because the copies do not share their estimates, each paying  $\log K/\eta$   
 293 over  $T/(d + 1)$  rounds. Sharing them is exactly the  $m = 1$  algorithm of Conjecture 4, whose analysis  
 294 is open.

295 **Lemma 7** (Coarsening). *For every play distribution and every  $g$ ,  $v_t(g) \leq v_t$ .*

296 *Proof.* It suffices to merge two states, the general case following by induction on the merges, and a  
 297 merged group of zero probability contributes nothing on either side by the convention above. Write  
 298  $P_1, P_2$  for the two columns,  $A_j = \sum_a x_t(a) P_j(a)^2$ ,  $B_j = \sum_a x_t(a) P_j(a)$  and  $\lambda = B_1/(B_1 + B_2)$ .  
 299 By Cauchy-Schwarz,  $(P_1 + P_2)^2 \leq \lambda^{-1} P_1^2 + (1 - \lambda)^{-1} P_2^2$  pointwise in  $a$ , so  $\sum_a x_t(a) (P_1 + P_2)^2 \leq$   
 300  $(B_1 + B_2)(A_1/B_1 + A_2/B_2)$ , and dividing by the merged denominator  $B_1 + B_2$  shows the merged  
 301 summand is at most  $A_1/B_1 + A_2/B_2$ , the two it replaces.  $\square$

302 *Proof of Theorem 2.* Running the algorithm on  $g$  is running it on the sensor  $(\mathcal{Z}, P^g)$ . The surrogate  
 303 losses below depend on  $x_t$  and so are predictable rather than oblivious, but the proof of Theorem 1  
 304 uses obliviousness nowhere, only that each  $c_t^g$  is  $\mathcal{F}_{t-1}$ -measurable given the play, so it applies un-  
 305 changed and bounds regret measured in the surrogate losses  $c_t^g(a) = \sum_z P^g(z | a) \bar{\theta}_t(z)$ , where  
 306  $\bar{\theta}_t(z) = \sum_{s: g(s)=z} q_t(s) \theta_t(s) / q_t^g(z)$  is the group mean the grouped estimator is unbiased for, by the  
 307 same computation as in Proposition 6 with  $S_t$  replaced by  $g(S_t)$ . Fix  $a$  and  $z$ . Both  $\{q_t(s)/q_t^g(z)\}$   
 308 and  $\{P(s | a)/P^g(z | a)\}$  are probability vectors on  $\{s : g(s) = z\}$ , so their  $\theta_t$ -averages lie in a  
 309 common interval of length  $\delta_t(g)$  and differ by at most  $\delta_t(g)$ . Weighting by  $P^g(z | a)$  and summing  
 310 over  $z$  gives  $|c_t^g(a) - c_t(a)| \leq \delta_t(g)$  for every  $a$ , hence  $\langle x_t, c_t \rangle - c_t(a^*) \leq \langle x_t, c_t^g \rangle - c_t^g(a^*) + 2\delta_t(g)$ .  
 311 Summing over  $t$  adds  $2 \sum_t \delta_t(g)$ , and Lemma 7 makes  $V(g)$  no larger than a bound valid for the  
 312 raw states.  $\square$

## 313 E An unknown action-to-state map

314 **The plug-in estimator.** The pairs  $(A_t, S_t)$  arrive without delay, so  $P$  is estimable from the  
 315 learner's own play while the outcomes are still in flight. Let  $N_a(t)$  count the rounds before  $t$  that  
 316 played  $a$  and  $N_{a,s}(t)$  those that also produced  $s$ , and set  $\hat{P}_t(s | a) = (N_{a,s}(t) + \alpha)/(N_a(t) + \alpha|S|)$   
 317 for a smoothing constant  $\alpha \in (0, 1]$  fixed independently of  $T$ , which is row-stochastic with every en-  
 318 try positive, so  $\hat{q}_t = x_t^\top \hat{P}_t$  is positive too, and both are  $\mathcal{F}_{t-1}$ -measurable. Smoothing moves an entry



from the empirical row by at most  $\alpha(|\mathcal{S}| - 1)/(N_a(t) + \alpha|\mathcal{S}|)$ , attained when every observation of  $a$  sits on one state, and moves the row in  $\ell_1$  by at most twice that. Both are of lower order than the sampling term at fixed  $\alpha$  and must be carried otherwise, and Appendix I shows the choice matters in practice. The plug-in estimate is  $\hat{c}_t(a) = \hat{P}_t(S_t | a)X_t/\hat{q}_t(S_t)$ . Write  $\varepsilon_t = \max_a \|\hat{P}_t(\cdot | a) - P(\cdot | a)\|_1$  for its accuracy and  $\rho_t = \max_{s: q_t(s) > 0} |\hat{q}_t(s) - q_t(s)|/q_t(s)$  for the relative error it induces on the state distribution, states of zero probability under the play being omitted since the estimator never charges them.

**Proposition 8.** *On  $\{\rho_t \leq \frac{1}{2}\}$  the plug-in estimate satisfies  $\hat{c}_t \geq 0$  and  $\sum_a x_t(a)\mathbb{E}[\hat{c}_t(a)^2 | \mathcal{F}_{t-1}] \leq 2\hat{v}_t \leq 2|\mathcal{S}|$ , and its conditional mean  $\bar{c}_t$  satisfies  $\sum_a x_t(a)(\bar{c}_t(a) - c_t(a)) = 0$  exactly, for every  $\hat{P}_t$  and with no hypothesis at all, and  $|\bar{c}_t(a) - c_t(a)| \leq 2\varepsilon_t(a) + 2\rho_t$  for every  $a$ , where  $\varepsilon_t(a) = \|\hat{P}_t(\cdot | a) - P(\cdot | a)\|_1$ .*

*Proof.* Write  $\Delta_t = \hat{P}_t - P$  and  $\delta_t = \hat{q}_t - q_t = \sum_a x_t(a)\Delta_t(\cdot | a)$ . Substituting these into  $\bar{c}_t(a) = \sum_s q_t(s)\hat{P}_t(s | a)\theta_t(s)/\hat{q}_t(s)$  and cancelling gives  $\bar{c}_t(a) - c_t(a) = \sum_s \theta_t(s)[q_t(s)\Delta_t(s | a) - P(s | a)\delta_t(s)]/\hat{q}_t(s)$ , so with  $\theta_t \in [0, 1]^{|\mathcal{S}|}$ ,

$$|\bar{c}_t(a) - c_t(a)| \leq \sum_s \frac{q_t(s)|\Delta_t(s | a)| + P(s | a)|\delta_t(s)|}{\hat{q}_t(s)}.$$

The play-weighted identity needs no hypothesis. By definition  $\sum_a x_t(a)\hat{P}_t(s | a) = \hat{q}_t(s)$ , so  $\sum_a x_t(a)\hat{c}_t(a) = X_t$  whatever  $\hat{P}_t$  is, and taking conditional expectations gives  $\sum_a x_t(a)\bar{c}_t(a) = \sum_s q_t(s)\theta_t(s) = \langle x_t, c_t \rangle$ . On  $\{\rho_t \leq \frac{1}{2}\}$  we have  $q_t \leq 2\hat{q}_t$ , and for the per-action bias the first sum of the display is at most  $2\|\Delta_t(\cdot | a)\|_1 = 2\varepsilon_t(a)$  and the second at most  $2\sum_s P(s | a)|\delta_t(s)|/q_t(s) \leq 2\rho_t$ , because  $P(\cdot | a)$  is a probability vector. The play-weighted identity does *not* survive reweighting: at  $x_t = (\frac{1}{2}, \frac{1}{2})$ ,  $P = I$ ,  $\hat{P} = \begin{pmatrix} .9 & .1 \\ .2 & .8 \end{pmatrix}$  and  $\theta_t = (0, 1)$  the bias is  $(\frac{1}{9}, -\frac{1}{9})$  with zero  $x_t$ -average, while weights  $(1, 0)$  leave  $\frac{1}{18}$ . For the second moment,  $\sum_a x_t(a)\mathbb{E}[\hat{c}_t(a)^2 | \mathcal{F}_{t-1}] \leq \sum_s \frac{q_t(s)}{\hat{q}_t(s)} \cdot \frac{\sum_a x_t(a)\hat{P}_t(s | a)^2}{\hat{q}_t(s)} \leq 2\hat{v}_t$ , and  $\hat{v}_t \leq |\mathcal{S}|$  by Proposition 6 applied to  $\hat{P}_t$ .  $\square$

**The warm-start guarantee.** WARM-START STATE-EXP3 plays uniformly for  $n_0$  rounds, forms  $\hat{P} = \hat{P}_{n_0+1}$  from those rounds alone, discards the cumulative estimates together with every outcome of a warm-start round that has not yet arrived, and then runs STATE-EXP3 with  $m = d + 1$ , that frozen  $\hat{P}$ , and each play mixed with the uniform action at rate  $\gamma$ . Freezing makes  $\hat{P}$  measurable with respect to the first phase, so  $\varepsilon$  and  $\rho$  are fixed before the second phase begins, and discarding is what keeps the first phase's inaccurate updates out of every later distribution.

**Theorem 9.** *Suppose  $\bar{p}(s) \geq 1/(\kappa|\mathcal{S}|)$  for every  $s$ , fix  $0 < \gamma \leq \frac{1}{2}$  and  $1 \leq d \leq T/2$ , and take  $n_0 \geq c\gamma^{-2}K\kappa^2|\mathcal{S}|^2 \log(K|\mathcal{S}|T)$  for a universal  $c$ . Then WARM-START STATE-EXP3 satisfies, for every  $\eta > 0$ ,*

$$\mathbb{E}[R_T] \leq n_0 + 1 + \gamma T + \frac{(d+1)\log K}{\eta} + 2\eta|\mathcal{S}|T + T(4\bar{\varepsilon} + 4\bar{\rho}),$$

where  $\bar{\varepsilon} = \tilde{O}(\sqrt{K|\mathcal{S}|/n_0})$  and  $\bar{\rho} = \tilde{O}(\kappa|\mathcal{S}|\sqrt{K/n_0}/\gamma) \leq \frac{1}{2}$  bound  $\varepsilon$  and  $\rho$  on an event of probability at least  $1 - 1/T$ . Tuning  $\eta$  and then  $\gamma \asymp (\kappa|\mathcal{S}|)^{1/2}(K/n_0)^{1/4}$  and  $n_0 \asymp T^{4/5}(\kappa|\mathcal{S}|)^{2/5}K^{1/5}$  gives  $\mathbb{E}[R_T] = \tilde{O}(\sqrt{d|\mathcal{S}|T \log K} + (\kappa|\mathcal{S}|)^{2/5}K^{1/5}T^{4/5})$ , provided  $T = \tilde{\Omega}(K(\kappa|\mathcal{S}|)^2)$ . That condition constrains  $T$  against  $K$ ,  $|\mathcal{S}|$  and  $\kappa$  only, so the hypothesis  $d \leq T/2$  is what carries the delay. Together they give  $n_0 \leq T/2 \leq T - d$ , and the same condition supplies the lower bound above with some  $\gamma \leq \frac{1}{2}$ . Outside that regime the tuning is vacuous and  $R_T \leq T$  is the statement.

*Proof.* The first phase has losses in  $[0, 1]$ , so it contributes at most  $n_0$ . Let  $G$  be the event that  $N_a(n_0 + 1) \geq n_0/(2K)$  for every  $a$ , that  $\|\hat{P}(\cdot | a) - P(\cdot | a)\|_1 \leq \bar{\varepsilon}$  for every  $a$ , and that  $\max_{s,a} |\hat{P}(s | a) - P(s | a)| \leq \bar{\rho}\gamma/(\kappa|\mathcal{S}|)$ . Uniform play gives  $\mathbb{E}[N_a] = n_0/K$ , so a multiplicative Chernoff bound, the  $\ell_1$  deviation bound for a multinomial, Hoeffding, and a union bound over the  $K|\mathcal{S}|$  entries put  $\Pr(G^c) \leq 1/T$  at the stated  $n_0$ , the pseudo-counts adding  $2\alpha(|\mathcal{S}| - 1)/(N_a + \alpha|\mathcal{S}|)$

to each  $\ell_1$  deviation and half that to each entry, of lower order at fixed  $\alpha$ . Off  $G$  bound the second phase by  $T$ , contributing at most 1.

On  $G$  the quantity  $\hat{P}$  is measurable with respect to the first phase, and mixing gives  $q_t(s) \geq \gamma \bar{p}(s) \geq \gamma/(\kappa|\mathcal{S}|)$  at every later round, so  $\rho_t \leq \kappa|\mathcal{S}| \max_{s,a} |\hat{P}(s|a) - P(s|a)|/\gamma \leq \bar{\rho} \leq \frac{1}{2}$  for every  $t > n_0$  and every realization of the second phase. Proposition 8 therefore applies at every such round, and conditionally on the first phase the second is exactly the algorithm of Theorem 1 driven by a nonnegative estimate with weighted second moment at most  $2\hat{v}_t \leq 2|\mathcal{S}|$ . Its proof gives  $\mathbb{E}[\sum_{t>n_0} (\langle p_t, \bar{c}_t \rangle - \bar{c}_t(a^*))] \leq (d+1) \log K/\eta + 2\eta|\mathcal{S}|T$ , the extra factor of two coming from  $p_t \leq x_t/(1-\gamma) \leq 2x_t$ . Playing  $x_t$  rather than  $p_t$  costs at most  $\gamma$  per round since  $c_t \in [0, 1]^K$ , so it remains to pass from  $\bar{c}_t$  to  $c_t$  under  $p_t$ . Proposition 8 gives  $\langle x_t, b_t \rangle = 0$  for the per-action bias  $b_t$ , and  $x_t = (1-\gamma)p_t + \gamma u$  for the uniform  $u$ , so  $\langle p_t, b_t \rangle = -\frac{\gamma}{1-\gamma} \langle u, b_t \rangle$ , which for  $\gamma \leq \frac{1}{2}$  is at most  $\max_a |b_t(a)|$  in absolute value. Each round therefore adds  $|\langle p_t, b_t \rangle| + |b_t(a^*)| \leq 2 \max_a |b_t(a)| \leq 4\bar{\varepsilon} + 4\bar{\rho}$ . Collecting the four contributions gives the display. For the rate, balancing  $\gamma T$  against  $2T\bar{\rho}$  gives the stated  $\gamma$  and a combined  $\tilde{O}(T(\kappa|\mathcal{S}|)^{1/2}(K/n_0)^{1/4})$ , and balancing that against  $n_0$  gives the stated  $n_0$ , under which  $T\bar{\varepsilon}$  and the constraint on  $n_0$  are of lower order.  $\square$

**Safe blending.** Blending the plug-in with the action-level estimate makes the guarantee safe. Let  $\lambda_t \in [0, 1]^K$  be  $\mathcal{F}_{t-1}$ -measurable and put  $\check{g}_t(a) = (1 - \lambda_t(a))\bar{c}_t(a) + \lambda_t(a)\hat{c}_t(a)$ .

**Proposition 10.**  $\check{g}_t \geq 0$ , its conditional mean satisfies  $\mathbb{E}[\check{g}_t(a) | \mathcal{F}_{t-1}] - c_t(a) = \lambda_t(a)(\bar{c}_t(a) - c_t(a))$  for the plug-in's conditional mean  $\bar{c}_t$  of Proposition 8, and

$$\sum_a x_t(a) \mathbb{E}[\check{g}_t(a)^2 | \mathcal{F}_{t-1}] \leq (K - \Pi_t) + \hat{V}_t,$$

for  $\Pi_t = \sum_a \lambda_t(a)$  and  $\hat{V}_t = \sum_a \lambda_t(a) x_t(a) \mathbb{E}[\hat{c}_t(a)^2 | \mathcal{F}_{t-1}]$ , which satisfies  $\hat{V}_t \leq 2\hat{v}_t$  on  $\{\rho_t \leq \frac{1}{2}\}$  and not in general, since the step  $q_t/\hat{q}_t \leq 2$  behind it is exactly that event. We write  $V_t^{\text{ub}}(\tau)$  below for a computable upper bound on  $\hat{V}_t$ , reserving  $\hat{V}_t$  for the conditional expectation itself.

*Proof.* Both parts are nonnegative and  $\lambda_t(a) \in [0, 1]$ . The mean identity is linearity together with  $\mathbb{E}[\bar{c}_t(a) | \mathcal{F}_{t-1}] = c_t(a)$ . For the second moment,  $u \mapsto u^2$  is convex, so  $\check{g}_t(a)^2 \leq (1 - \lambda_t(a))\bar{c}_t(a)^2 + \lambda_t(a)\hat{c}_t(a)^2$ , and  $x_t(a) \mathbb{E}[\bar{c}_t(a)^2 | \mathcal{F}_{t-1}] \leq 1$  by Proposition 6.  $\square$

So a weight of  $\lambda_t(a)$  on the plug-in buys a variance saving of  $\lambda_t(a)$  and pays a bias of  $\lambda_t(a)$  times the plug-in's, and both endpoints are exact. Let  $r_t(a)$  and  $\hat{\rho}_t$  be any sequences computable from the immediate pairs that are *simultaneously valid*, meaning  $\varepsilon_t(a) \leq r_t(a)$  for every  $a$  and  $\rho_t \leq \hat{\rho}_t$  at every  $t \leq T$ , on an event of probability at least  $1 - 1/T$ . Write  $V_t^{\text{ub}}(\tau) = 2 \sum_{a: r_t(a) \leq \tau} x_t(a) \sum_s \hat{P}_t(s|a)^2/\hat{q}_t(s)$ , which the learner can evaluate and which dominates  $\hat{V}_t$  on  $\{\rho_t \leq \frac{1}{2}\}$ . SAFE-POOL STATE-EXP3 sets  $\lambda_t \equiv 0$  when  $\hat{\rho}_t > \frac{1}{2}$ , and otherwise picks the best member of the candidate set  $\{\perp\} \cup \{r_t(a) : a \leq K\}$ , where  $\perp$  stands for  $\lambda_t \equiv 0$  and a numerical candidate  $\tau$  stands for  $\lambda_t(a) = \mathbf{1}\{r_t(a) \leq \tau\}$ , best meaning smallest

$$\Psi_t(\tau) = \frac{\eta}{2} [(K - \Pi_t(\tau)) + V_t^{\text{ub}}(\tau)] + (4\tau + 4\hat{\rho}_t) \mathbf{1}\{\Pi_t(\tau) > 0\}, \quad \Psi_t(\perp) = \frac{\eta}{2} K.$$

Write  $\Psi_t^*$  for that smallest value. The candidate  $\perp$  is listed apart from the numerical thresholds because a valid radius may equal zero, in which case  $\tau = 0$  pools the actions attaining it rather than abstaining, so  $\perp$  rather than  $\tau = 0$  is what keeps  $\frac{\eta}{2} K$  available on every round.

**Theorem 11.** For every  $\eta > 0$ , SAFE-POOL STATE-EXP3 with  $m = d + 1$  satisfies

$$\mathbb{E}[R_T] \leq 1 + \frac{(d+1) \log K}{\eta} + \mathbb{E}\left[\sum_{t=1}^T \Psi_t^*\right],$$

so  $\eta = \sqrt{2(d+1) \log K/(KT)}$  gives  $\mathbb{E}[R_T] \leq 1 + \sqrt{2(d+1)KT \log K}$  on every instance, which is the guarantee of action-level exponential weights, with an improvement on each round whose  $\Psi_t^*$  is strictly below  $\frac{\eta}{2} K$ . The certificate is computed from the learner's own observations and so is random, which is why the display carries an expectation. The tuned bound does not, because  $\Psi_t^* \leq \frac{\eta}{2} K$  holds on every path.

404 *Proof.* On the validity event, of probability at least  $1 - 1/T$  and contributing at most 1 off it, pooling  
405 happens only when  $\hat{\rho}_t \leq \frac{1}{2}$  and hence only when  $\rho_t \leq \frac{1}{2}$ , which is what makes  $V_t^{\text{ub}}$  dominate the  
406 conditional second moment. Proposition 10 then supplies a nonnegative estimate whose weighted  
407 second moment is at most  $(K - \Pi_t) + \hat{V}_t$ , so the argument of Appendix D applies to the sur-  
408rogate losses. Passing to  $c_t$  adds both the played bias  $|\sum_a x_t(a)\lambda_t(a)b_t(a)|$  and the comparator  
409 bias  $\lambda_t(a^*)|b_t(a^*)|$ , for  $b_t = \bar{c}_t - c_t$ . The cancellation of Proposition 8 is lost under per-action  
410 weights, as the counterexample there shows, so each is bounded separately. On a pooled action  
411  $\varepsilon_t(a) \leq r_t(a) \leq \tau$  gives  $|b_t(a)| \leq 2\tau + 2\hat{\rho}_t$ , and since  $\sum_a x_t(a)\lambda_t(a) \leq 1$  and  $\lambda_t(a^*) \leq 1$  the two  
412 together are at most  $4\tau + 4\hat{\rho}_t$ , charged only when something is pooled. The second-moment bound  
413 and the two bias terms are all functions of the observations, so the per-round costs they assemble  
414 into are the random  $\Psi_t^*$  and the sum is taken in expectation. For the tuned bound,  $\perp$  belongs to the  
415 candidate set and  $\Psi_t(\perp) = \frac{\eta}{2}K$ , so  $\Psi_t^* \leq \frac{\eta}{2}K$  on every path whatever the radii are.  $\square$

416 **Why certification stays conservative.** The guarantee is safe and the algorithm is inert. Pooling  
417 is certified when  $4(\tau + \hat{\rho}_t) \leq \frac{\eta}{2}(\Pi_t - V_t^{\text{ub}})$ . At the tuned  $\eta$  both sides fall as  $T^{-1/2}$ , the left through  
418  $\tau \asymp \sqrt{K \log(K|\mathcal{S}|T)/T}$  under uniform play and the right through  $\eta \asymp \sqrt{\log K/(KT)}$ . Their ratio  
419 is  $\Theta(\sqrt{\log(K|\mathcal{S}|T)/\log K})$ , which exceeds one and does not vanish with the horizon, growing  
420 slowly in  $T$  through the logarithmic factors, so no horizon makes the certificate clear. Appendix I  
421 reports behavior consistent with this explanation, the algorithm reproducing action-level regret to  
422 the digit with a pooled fraction of zero.

423 The obstruction is not the shape of the bias bound. Regret charges  $\langle x_t, b_t \rangle - b_t(a^*)$  for the per-action  
424 bias  $b_t$ , while Proposition 8 charges  $\max_a |b_t(a)|$ . Measured against the budget itself at  $K = 40$ ,  
425  $|\mathcal{S}| = 8$ ,  $T = 8000$  and  $d = 50$ , the first is 0.42 times it and the second 1.67 times it, so on that  
426 trajectory the realized bias lies below the best-case variance budget and a tighter bias bound would  
427 close the analytical gap. Two exact refinements fail to deliver one. Both  $\sum_s \Delta_t(s | a)$  and  $\sum_s \delta_t(s)$   
428 vanish, so a constant may be subtracted and a maximum replaced by an oscillation, and Cauchy-  
429 Schwarz against  $q_t$  replaces the comparator term by  $\sqrt{\nu_t(a^*)} \chi_t$  for its own effective dimension  $\nu_t$   
430 and a chi-square  $\chi_t$  between  $\hat{q}_t$  and  $q_t$ . Measured, these are 6.1 and 63 times the truth against 3.9  
431 for the maximum they replace.

432 What the algorithm can verify from its own counts is 104 times the budget, and Bernstein radii do not  
433 improve on Hoeffding ones (Appendix I), so the barrier is certifying  $\rho_t$ , the relative error the estimate  
434 induces on the state distribution. The play-weighted identity of Proposition 8 says the whole charge  
435 is the comparator's, and it routes through  $\rho_t$  only because  $\hat{q}_t = x_t^\top \hat{P}_t$  predicts the state distribution  
436 out of every row of  $P$ . Predicting is expensive for that reason, needing  $t = \Omega(K|\mathcal{S}| \log(K|\mathcal{S}|T)/\rho^2)$   
437 rounds, and the resulting certificate clears for no  $K \leq T$ , since it would need  $\log K \geq 16c|\mathcal{S}|$  for  
438  $c = \log(2|\mathcal{S}|T)$ , which  $\log K \leq \log T$  cannot meet.

439 Counting the states costs less and still buys nothing. With  $x_t$  frozen over a window the empirical  
440 state frequency is a plain multinomial estimate of  $q_t$ , needing  $n = \Omega(|\mathcal{S}| \log(|\mathcal{S}|T)/\rho^2)$  samples, a  
441 factor  $K$  fewer than predicting and free of  $P$ , and the certificate does clear once  $K \log K \geq 64c|\mathcal{S}|$ ,  
442 about  $K \geq 10^3$  at  $|\mathcal{S}| = 8$ . The window that buys it must satisfy  $n \geq 32c|\mathcal{S}|T/(K \log K)$ , a  
443 constant fraction of the horizon, and freezing the play over blocks of length  $B$  costs  $\sqrt{2TBK \log K}$ ,  
444 which at that  $B$  is  $T\sqrt{128c|\mathcal{S}|}$  and so linear in  $T$ . Both routes fail for one reason: the certificate  
445 must resolve  $q_t$  to relative accuracy  $\Theta(\sqrt{K \log K/T})$ , finer than either estimate delivers from the  
446 data the learner has.

447 In words, an unknown action-to-state matrix costs a bias in two accuracies, the  $\ell_1$  error of  $\hat{P}$  and  
448 the relative error it induces on the state distribution, and only the second needs forced exploration,  
449 which is what sets the rate. **The bound does not certify an improvement from the channel it**  
450 **analyzes.** Its second term exceeds  $\sqrt{KT}$  at every horizon with  $T > K$ , by a factor  $(T/K)^{3/10}$ ,  
451 so a learner holding this guarantee alone should run action-level exponential weights instead. The  
452 experiments suggest the analysis is conservative rather than the algorithm weak, since Appendix I  
453 shows it tracking known  $P$  closely, and sharpening it is the natural next result.

## F Feedback models, and what transfers

Write  $\Lambda_2 = \sum_t \min\{1, \|c_t - m_t\|_2^2\}$  for the inaccuracy measure of Bar-On and Mansour (2025). Unlike  $E_2$ , which charges only the worst action in a round,  $\Lambda_2$  grows with how many actions are predicted wrongly, and since  $\|v\|_\infty \leq \|v\|_2 \leq \sqrt{q}\|v\|_\infty$  for a vector  $v$  with  $q$  nonzero coordinates,  $E_2 \leq \Lambda_2 \leq q E_2$  with both ends attained.

Bar-On and Mansour (2025) analyze a model in which the played coordinate of the loss vector is observed exactly. Section 3 instead observes a random outcome whose conditional mean is the loss. The following records precisely what survives the substitution.

**Proposition 12.** *Take the action-level  $\tilde{c}_t(a) = \mathbf{1}\{A_t = a\}X_t/x_t(a)$  of Proposition 6 under the model of Section 3. Then for every action  $a$ , (i)  $\mathbb{E}[\tilde{c}_t(a) \mid \mathcal{F}_{t-1}] = c_t(a)$ , (ii)  $0 \leq \tilde{c}_t(a) \leq 1/x_t(a)$ , and (iii)  $\mathbb{E}[\tilde{c}_t(a)^2 \mid \mathcal{F}_{t-1}] \leq 1/x_t(a)$ . The centered estimator  $\check{c}_t(a) = m_t(a) + \mathbf{1}\{A_t = a\}(X_t - m_t(a))/x_t(a)$  also satisfies (i), and its residual obeys  $\mathbb{E}[(\check{c}_t(a) - m_t(a))^2 \mid \mathcal{F}_{t-1}] \leq 1/x_t(a)$ .*

*Proof.* For (i), by the tower rule  $\mathbb{E}[\mathbf{1}\{A_t = a\}X_t \mid \mathcal{F}_{t-1}] = x_t(a) \mathbb{E}[X_t \mid A_t = a] = x_t(a)c_t(a)$ , and dividing by  $x_t(a)$  gives the claim. Item (ii) holds because  $X_t \in [0, 1]$ . For (iii),  $\mathbb{E}[\tilde{c}_t(a)^2 \mid \mathcal{F}_{t-1}] = \mathbb{E}[X_t^2 \mathbf{1}\{A_t = a\}]/x_t(a)^2 = \mathbb{E}[X_t^2 \mid A_t = a]/x_t(a) \leq 1/x_t(a)$  because  $X_t \leq 1$ . The statements for  $\check{c}_t$  follow the same computation with  $X_t$  replaced by  $X_t - m_t(a)$ , using  $|X_t - m_t(a)| \leq 1$ .  $\square$

In words, replacing an exactly observed loss by a bounded unbiased draw of it changes neither the mean of the estimator nor the moment bounds an analysis consumes.

Under exact feedback the corresponding estimator satisfies (i) to (iii) with identical bounds, since  $c_t(a) \leq 1$ . Consequently any step of an exact-feedback analysis that uses the observation only through unbiasedness and these two moment bounds is unchanged under noisy outcomes. What this does not cover is the comparison between the estimate built from the stale action-loss vector and the estimate built from the true loss. Under exact feedback that difference is determined by the played action. Under noisy outcomes it carries an additional mean-zero term of magnitude up to  $1/x_t(A_t)$ , and bounding its cumulative effect over a window of  $d$  rounds is what requires the stability property falsified in Appendix H. An upper bound for the noisy model is Theorem 1. What remains open is one adaptive to  $E_2$ .

## G Proofs

The measure  $E_2$  is controlled by the state-loss variation  $W$ , and no converse bound holds in general.

**Lemma 13** (Delay window).  *$E_2 \leq d^2 W$  for  $W = \sum_t \|\theta_t - \theta_{t-1}\|_\infty^2$ . The ratio  $E_2/(d^2 W)$  tends to one when  $\theta$  drifts monotonically in one coordinate at a constant rate and  $T/d \rightarrow \infty$ , the first  $d$  rounds contributing  $\sum_{j < d} j^2$  rather than  $d^2$  each under the convention  $m_t = c_1$ .*

In words, accumulated squared prediction error is at most the state-loss variation inflated by the square of the delay, and no better bound in  $W$  alone is available.

*Proof of Lemma 13.* Throughout write  $\Delta_j = \theta_j - \theta_{j-1}$ , set  $\theta_0 = \theta_1$ , so  $\Delta_1 = 0$ , and read  $c_r$  as  $c_1$  for  $r \leq 0$ , matching the convention  $m_t = c_1$  for  $t \leq d$  of Section 3. Then  $c_t - c_{t-d} = P \sum_{j=t-d+1}^t \Delta_j$ . Because the rows of  $P$  are probability vectors, averaging cannot increase the supremum norm, so  $\|c_t - c_{t-d}\|_\infty \leq \sum_j \|\Delta_j\|_\infty$ . By Cauchy–Schwarz applied to the  $d$  summands,  $\|c_t - c_{t-d}\|_\infty^2 \leq d \sum_{j=t-d+1}^t \|\Delta_j\|_\infty^2$ . Summing over  $t$  and using that each increment belongs to at most  $d$  windows gives  $E_2 \leq d^2 W$ . The ratio  $E_2/(d^2 W)$  approaches one when every increment has the same magnitude, sign and active coordinate, and some action reaches that coordinate deterministically, since then no cancellation occurs within a full window and  $P$  passes the increment through undiminished. Without the second condition the drift can lie in the kernel of  $P$  and the ratio can be zero.  $\square$

**Lemma 14** (Information isolation). *If  $h \leq d$  then  $A_t$  is independent of the signs governing its own block, and this remains true when the learner is provided with the exact stale action-loss vector  $m_t$ .*

500 *Proof of Lemma 14.* For  $t$  in block  $b$ , every realized outcome usable when choosing  $A_t$  comes from  
 501 a round  $r \leq t - d - 1$ . Since  $t \leq bh$  and  $h \leq d$ , we have  $r \leq bh - d - 1 < (b - 1)h$ , so every usable  
 502 realized outcome depends only on signs from blocks strictly before  $b$ , whose joint law depends only  
 503 on  $\sigma_2, \dots, \sigma_{b-1}$ . Hence by conditional independence  $A_t$  is independent of block  $b$ 's signs. For the  
 504 oracle, take any  $h \leq d$ . If  $t > d$  then  $t - d \leq bh - d \leq (b - 1)h$  by the same computation,  
 505 so  $m_t = c_{t-d}$  is measurable with respect to the signs of blocks strictly before  $b$ . If  $t \leq d$  then  
 506  $m_t = c_1 = \frac{1}{2}\mathbf{1}$ , which is deterministic because block 1 is neutral. In both cases  $m_t$  is independent  
 507 of block  $b$ 's signs, so adjoining it to the learner's information leaves the conclusion intact.  $\square$

508 The construction in full. Take states  $s_0, \dots, s_q$ , a stationary deterministic map sending  $a_j$  to  $s_j$  for  
 509  $j \leq q$  and every other action to  $s_0$ , and independent Rademacher signs  $\sigma_b^{(j)}$  for  $2 \leq b \leq B = \lfloor T/h \rfloor$ .  
 510 Set  $\theta_t(s_0) = \frac{1}{2}$  throughout,  $\theta_t \equiv \frac{1}{2}$  on block 1, and  $\theta_t(s_j) = \frac{1}{2} - \varepsilon \sigma_b^{(j)}$  on each later block  $b$ . The  
 511 neutral first block makes  $c_1 = \frac{1}{2}\mathbf{1}$ , so the convention  $m_t = c_1$  for  $t \leq d$  of Section 3 supplies a  
 512 constant over exactly the rounds that precede any usable feedback, which is what Lemma 14 needs  
 513 at  $b = 1$ . Rounds after the last complete block carry no drift, costing a constant factor.

514 **Lemma 15** (Rademacher lower tail). *Let  $S_B$  be a sum of  $B \geq 32$  independent Rademacher signs.*  
 515 *For every  $1 \leq u \leq \sqrt{B}/32$ ,  $\Pr(S_B \geq u\sqrt{B}) \geq \frac{u}{4}e^{-64u^2}$ .*

516 *Proof.* Write  $S_B = 2X - B$  with  $X \sim \text{Bin}(B, \frac{1}{2})$ , so that  $\{S_B \geq u\sqrt{B}\} = \{X \geq B/2 + u\sqrt{B}/2\}$ .  
 517 Passing to  $X$  removes the lattice gap, since  $X$  takes every integer value in  $[0, B]$  whereas  $S_B$  takes  
 518 only those of the parity of  $B$ . Put  $n = \lfloor B/2 \rfloor$  and  $t = \lceil u\sqrt{B}/2 \rceil + 1$ , so  $X \geq n + t$  implies the  
 519 event, and  $t \leq u\sqrt{B}$  because  $u\sqrt{B} \geq \sqrt{32} > 4$ .

520 For  $j \geq 1$  the ratio of consecutive binomial probabilities is  $\Pr(X = n + j)/\Pr(X = n + j - 1) = (B - n - j + 1)/(n + j)$ . Using  $(B - 1)/2 \leq n \leq B/2$ , its distance from one is at  
 521 most  $(2j - 1)/(n + j) \leq (4j - 2)/(B - 1) \leq 8j/B$  for  $B \geq 2$ . Restricting to  $j \leq B/16$  keeps  
 522  $8j/B \leq \frac{1}{2}$ , where  $\log(1 - x) \geq -2x$  holds, so summing over  $i \leq j$  gives  $\Pr(X = n + j) \geq \Pr(X = n)e^{-16j(j+1)/(2B)} \geq \Pr(X = n)e^{-16j^2/B}$ . The central term satisfies  $\Pr(X = n) \geq 1/(2\sqrt{B})$  for  
 523 every  $B \geq 2$ .

524 Sum over the  $t$  integers  $j \in \{t, \dots, 2t - 1\}$ , all of which are at most  $2t$  and hence at most  $B/16$   
 525 because  $u \leq \sqrt{B}/32$  forces  $2t \leq 2u\sqrt{B} \leq B/16$ . Each term is at least  $e^{-64t^2/B}/(2\sqrt{B})$ , and  
 526  $t \leq u\sqrt{B}$  makes the exponent at most  $64u^2$ , so  $\Pr(X \geq n + t) \geq te^{-64u^2}/(2\sqrt{B}) \geq \frac{u}{4}e^{-64u^2}$   
 527 using  $t \geq u\sqrt{B}/2$ .  $\square$

530 Three regimes give the maximal inequality the lower bound needs. Each  $S_B^{(j)}$  is sub-Gaussian  
 531 with variance proxy  $B$ , so  $\mathbb{E} \max_j (S_B^{(j)})^+ \leq \sqrt{2B \log q} + \sqrt{B}$  throughout. For the matching  
 532 lower bound, first suppose  $\log q \leq B/8$  and  $\log q \geq 128$ . Taking  $u = \sqrt{\log q/128}$ , which  
 533 then lies in  $[1, \sqrt{B}/32]$ , makes  $e^{-64u^2} = q^{-1/2}$ , so  $\Pr(S_B^{(j)} \geq u\sqrt{B}) \geq \frac{u}{4}q^{-1/2}$  and inde-  
 534 pendence across  $j$  gives  $\Pr(\max_j S_B^{(j)} < u\sqrt{B}) \leq e^{-u\sqrt{q}/4}$ , at most  $\frac{1}{2}$  beyond an absolute  $q$ .  
 535 Hence  $\mathbb{E} \max_j (S_B^{(j)})^+ \geq \frac{1}{2}u\sqrt{B} = \Omega(\sqrt{B \log q})$ . Second, if  $\log q > B/8$  and  $B \geq 1024$ , take  
 536  $u = \sqrt{B}/32$ , so  $u\sqrt{B} = B/32$  and  $e^{-64u^2} = e^{-B/16}$ , and  $q \geq e^{B/8}$  makes  $qe^{-B/16}$  diverge,  
 537 giving  $\mathbb{E} \max_j (S_B^{(j)})^+ = \Omega(B)$ , which is  $\Omega(\sqrt{B \min\{1 + \log q, B\}})$  in that regime. Third, when  
 538  $\log q < 128$  or  $B < 1024$  the quantity  $\sqrt{B \min\{1 + \log q, B\}}$  is  $O(\sqrt{B})$ , and a single walk already  
 539 gives  $\mathbb{E}(S_B^{(1)})^+ = \frac{1}{2}\mathbb{E}|S_B| \geq \frac{1}{2}(\mathbb{E}S_B^2)^{3/2}(\mathbb{E}S_B^4)^{-1/2} \geq \frac{1}{2}\sqrt{B/3}$  by Hölder, since  $\mathbb{E}S_B^2 = B$  and  
 540  $\mathbb{E}S_B^4 \leq 3B^2$ . Combining the three regimes,  $\mathbb{E} \max_j (S_B^{(j)})^+ = \Theta(\sqrt{B \min\{1 + \log q, B\}})$ . This is  
 541 the route by which maxima of independent random walks fix the minimax rate for prediction with  
 542 expert advice (Cesa-Bianchi and Lugosi, 2006).

543 **Proof idea.** Divide time into blocks of length  $d$  and give  $q$  actions private states whose losses are  
 544 perturbed by independent Rademacher signs within each block. Every observation available when  
 545 an action is chosen comes from an earlier block, and the supplied  $m_t = c_{t-d}$  depends only on  
 546 earlier signs, so the learner cannot predict the signs governing its current block. Its expected loss is

therefore  $T/2$ , while the best fixed action in hindsight benefits from the maximum of  $q$  independent random walks. Choosing the amplitude to make  $E_2 \leq \mathcal{E}$  gives the stated scale.

*Proof of Theorem 3.* Take  $h = d$  and  $\varepsilon = \frac{1}{2}\sqrt{\mathcal{E}/T}$ . By Lemma 14 every algorithm's expected loss equals  $T/2$ , since its action is independent of the current block's signs and all states have mean  $\frac{1}{2}$  marginally. Its regret is therefore exactly the comparator's fluctuation advantage,  $\varepsilon d \mathbb{E}[\max_{j \leq q} (\sum_{2 \leq b \leq B} \sigma_b^{(j)})^+]$  with  $B = \lfloor T/d \rfloor$ , over the  $B - 1$  randomized blocks. Per round  $\|c_t - m_t\|_\infty \leq 2\varepsilon$ , because the supremum norm charges only the largest coordinate and each coordinate moves by at most  $2\varepsilon$  at a block boundary; the gap is  $\varepsilon$  across the first randomized block, where  $m_t$  is still the neutral  $c_1$ , and lies in  $\{0, 2\varepsilon\}$  thereafter. Hence  $E_2 \leq 4\varepsilon^2 T = \mathcal{E}$  holds deterministically. The budget must bind on the realized sequence, since a budget holding only in expectation would not constrain the instance selected by Yao's principle. By Lemma 15 and the discussion following it, the expected maximum of the positive parts of  $q$  independent Rademacher sums of length  $B - 1$  is  $\Theta(\sqrt{(B - 1) \min\{1 + \log q, B - 1\}})$ , which covers both regimes and the case  $q = 1$ . The hypothesis  $d \leq T/2$  gives  $B \geq 2$  and hence  $B - 1 \geq B/2$ , so replacing  $B - 1$  by  $B$  costs a factor at most  $\sqrt{2}$  in each branch of the minimum, including the saturated branch where the minimum is  $B - 1$  rather than  $1 + \log q$ . Substituting  $B = \lfloor T/d \rfloor$  and  $\varepsilon$  therefore gives  $\Theta(\sqrt{d\mathcal{E} \min\{1 + \log q, T/d\}})$ . Rounds after the last complete block carry no drift and change the constant only. When the oracle is withheld, the  $\sqrt{KT}$  term is Esposito et al. (2023, Thm. 5.1), whose construction is stationary and therefore lies in the present model class.  $\square$

**Proposition 16.** Suppose  $X_t = c_t(A_t)$ . On instances whose drift is carried by a single action,  $\Lambda_2 = E_2$ , so the bound of Bar-On and Mansour (2025) reads  $\tilde{O}(\sqrt{KT} + d E_2)$  and its drift contribution  $\sqrt{d E_2}$  matches Theorem 3 up to logarithmic factors.

In words, single-action drift makes the Euclidean and supremum measures agree, so on that family the drift term of the published upper bound is unimprovable in order. Its  $\sqrt{KT}$  term is not matched because the oracle supplies an entire stale loss vector, so the oracle-assisted problem need not retain bandit exploration complexity. Neither result provides an  $E_2$ -adaptive upper bound under noisy outcomes.

*Proof of Proposition 16.* If only one coordinate of  $c_t - m_t$  is nonzero then  $\|c_t - m_t\|_2 = \|c_t - m_t\|_\infty$ . On the family of Section 6 that coordinate has magnitude at most  $2\varepsilon \leq 1$ , so the truncation at one in  $\Lambda_2$  is inactive and  $\Lambda_2 = E_2$ . The bound of Bar-On and Mansour (2025) then reads  $\tilde{O}(\sqrt{KT} + d E_2)$ , whose  $\sqrt{d E_2}$  contribution Theorem 3 matches up to the logarithmic factor at  $q = 1$ . Its  $\sqrt{KT}$  contribution is not matched.  $\square$

## H The two-action counterexample

Take  $K = 2$  with the hybrid regularizer of Zimmert and Seldin (2020), Tsallis parameter  $\beta = 0.01$ , learning rate  $\eta = 0.5$  and residual scale  $C = 1$ . The iterate solves the stationarity condition  $g(x_1) + L_1 = g(x_2) + L_2$  with  $x_1 + x_2 = 1$ , where

$$g(x) = -\frac{1}{2\beta}x^{-1/2} + \frac{1}{\eta}\log x$$

and  $L_a$  is the cumulative estimate for action  $a$ . Suppose the cumulative estimates place action 1 at probability  $x_1$ . A single maximally negative centered residual  $-C/x_1$  is then delivered to action 1. This is realizable in the model of Section 3, since  $m_t(1) = 1$  and  $X_t = 0$  give exactly  $\tilde{c}_t(1) - m_t(1) = -C/x_1$  with  $C = 1$ . Solving the stationarity condition again gives the following.

| $x_1$ before          | $x_1$ after           | ratio  |
|-----------------------|-----------------------|--------|
| $9.15 \times 10^{-3}$ | $1.46 \times 10^{-2}$ | 1.6    |
| $9.83 \times 10^{-4}$ | $7.39 \times 10^{-3}$ | 7.5    |
| $2.19 \times 10^{-4}$ | $9.98 \times 10^{-1}$ | 4561   |
| $5.80 \times 10^{-5}$ | 1.00                  | 17226  |
| $4.86 \times 10^{-6}$ | 1.00                  | 205641 |

588 The transition sits at  $4\beta^2 C^2 = 4 \times 10^{-4}$ , which is where the Tsallis term  $x^{-1/2}$  ceases to dominate  
589 the entropy term  $\log x$  in  $g$ , so the one-step linearization underlying a constant-factor bound fails.  
590 The violated statement is the local-stability inequality asserting that  $x_{t'}(a)/x_t(a)$  stays bounded by  
591 a constant for all  $t'$  within  $d$  rounds of  $t$ . A single round refutes it, so no accumulation argument  
592 is needed. This invalidates that lemma and the proof route resting on it, and not every argument  
593 that might use the same regularizer. It does not refute Conjecture 4, which concerns the regret rate  
594 and may be provable by other means. Devices that bound the residual magnitude introduce either a  
595 first-order bias dominating the target term or a probability floor narrowing the delay regime.

## 596 I Numerical detail

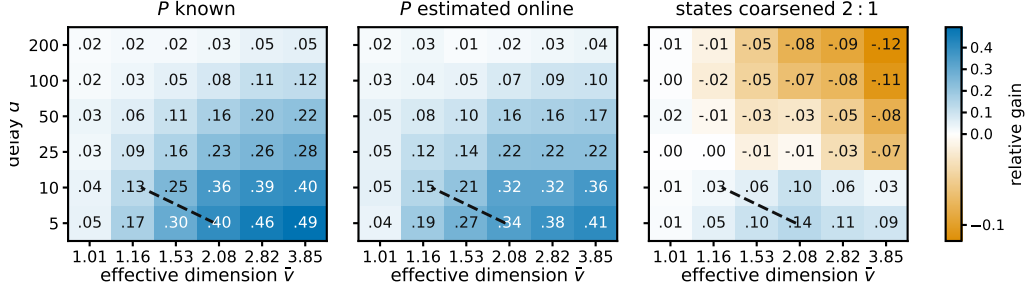


Figure 2: Relative gain  $(R_{\text{action}} - R_{\text{state}})/R_{\text{action}}$  of pooling through the state, for the practical  $m = 1$  variant, at  $K = 40$ ,  $|\mathcal{S}| = 8$ ,  $T = 5000$  and 12 paired seeds. Blue is a gain, and  $\hat{v}$  denotes a Monte Carlo estimate of  $\hat{v} = \sup_x v(x)$  obtained from 4000 sampled play distributions of the *raw* map, which for the coarsened panel is not the  $\bar{v}(P^g)$  that Theorem 2 charges. The dashed line marks where the bound of Theorem 1, which analyzes  $m = d + 1$ , falls below  $\tilde{O}(\sqrt{(K + d)T})$ . It compares two upper bounds for a variant other than the one plotted, so it is a reference and not a prediction.

597 Fifteen studies, alongside Figure 2, which sweeps row overlap against delay in the three settings the  
598 studies treat separately,  $P$  known,  $P$  estimated, and states coarsened. Studies one to thirteen draw  $P$   
599 from a Dirichlet and studies fourteen and fifteen do not. Study fifteen is the one where STATE-EXP3  
600 is beaten. Each study is reported as an observation, an interpretation, and a limitation. Studies 8  
601 and 9 ask whether the two theorems describe the right *dependence* rather than whether a bound holds,  
602 which is the question a loose bound leaves open. Study 10 is a control for study 6 and reverses its  
603 conclusion, and study 13 reports a rule that does not work. All policies run on the same environment  
604 parameters and common random-number streams, so a seed fixes everything except which estimator  
605 is formed, and differences are taken within a seed before averaging. Round  $r$ 's outcome is consumed  
606 before round  $r + d + 1$ , matching Section 3. Reported  $\pm$  is one standard error of the paired difference.  
607 Throughout this appendix, *sample-path pseudo-regret* means  $\sum_t c_t(A_t) - \min_a \sum_t c_t(a)$ , evaluated  
608 on one realization of the learner and environment randomness. Averaging it over seeds estimates the  
609 expected pseudo-regret that the theorems bound. We use that name rather than “realized regret” to  
610 distinguish it from regret computed from the sampled outcomes  $X_t$ , which is not what these studies  
611 report.

612 **Does the bound hold, and does the estimator help?** With  $|\mathcal{S}| = 4$ ,  $T = 10000$ ,  $d = 50$ , drift band  
613 0.05 and 100 seeds:

| $K$ | state, $m = d + 1$ | state, $m = 1$ | action, $m = 1$ | paired gain     | state better in |
|-----|--------------------|----------------|-----------------|-----------------|-----------------|
| 8   | 455.1              | 368.2          | 387.6           | $19.3 \pm 3.4$  | 73/100          |
| 20  | 618.3              | 493.7          | 547.5           | $53.8 \pm 5.6$  | 87/100          |
| 40  | 734.3              | 574.5          | 696.1           | $121.5 \pm 9.4$ | 99/100          |
| 80  | 780.4              | 614.8          | 784.0           | $169.3 \pm 9.9$ | 100/100         |

614 Observed. Sample-path pseudo-regret for the analyzed variant stayed below  $\sqrt{2(d + 1)|\mathcal{S}|T \log K}$   
615 in 100 of 100 seeds at every  $K$ , the bound running from 2913 to 4228 against realized 455 to 780.

617 The plain variant beat the action-level baseline at every  $K$ , by a margin growing from 19 to 169 as  
618  $K$  rose tenfold with  $|\mathcal{S}|$  fixed, and it beat the analyzed variant throughout. Sweeping the delay at  
619  $K = 40$  gave state-pooled 387, 569 and 731 against action-level 653, 699 and 773 at  $d = 10, 50$   
620 and 200.

621 Interpretation. The margin growing in  $K$  at fixed  $|\mathcal{S}|$  is the prediction that separates the two estima-  
622 tors, so the improvement is attributable to the state channel the effective dimension attributes to it.  
623 Interleaving looks like a price of the analysis rather than of the estimator. The narrowing of the gap  
624 as the delay grows is what an additive delay term predicts and a multiplicative one does not, which  
625 is the diagnostic evidence behind Conjecture 4.

626 Limitation. This is one synthetic family with a fixed drift band, the plain variant carries no guarantee,  
627 and a simulation cannot establish that a bound holds, only that sample-path pseudo-regret stayed  
628 below it in the runs performed.

629 **What does an unknown map cost?** What follows evaluates a practical variant, the plug-in estimate  
630 with  $\hat{P}_t$  updated every round, no restart,  $m = 1$  and  $\alpha = 0.01$ , and not the warm-start algorithm  
631 of Theorem 9, which freezes  $\hat{P}$  and restarts. Against the same algorithm given  $P$  and against the  
632 action-level baseline, over 60 paired seeds and at mixing rate  $\gamma$ :

| $K$ | $\gamma$ | known $P$ | plug-in | action-level | plug-in gain     |
|-----|----------|-----------|---------|--------------|------------------|
| 8   | 0.00     | 366.2     | 365.3   | 385.4        | $20.0 \pm 4.3$   |
| 20  | 0.00     | 480.3     | 477.9   | 529.0        | $51.2 \pm 7.6$   |
| 40  | 0.00     | 582.8     | 585.3   | 710.2        | $124.8 \pm 11.2$ |
| 80  | 0.00     | 625.2     | 623.1   | 802.2        | $179.1 \pm 12.9$ |
| 40  | 0.02     | 590.5     | 592.8   | 712.9        | $120.1 \pm 10.4$ |
| 40  | 0.10     | 618.9     | 623.5   | 735.5        | $112.0 \pm 10.1$ |

634 Observed. The plug-in landed within 0.5 per cent of known  $P$  at every  $K$  and beat action-level  
635 weighting by 20 to 179 as  $K$  rose tenfold. Forced exploration did not help,  $\gamma = 0$  being best at every  
636  $K$ . At  $K = 40$  the sampled maps have  $\kappa \approx 1.14$ , so the exploration term  $(\kappa|\mathcal{S}|)^{2/5}K^{1/5}T^{4/5}$  of  
637 Theorem 9 evaluates to about 6100 and its leading term to about 2700, against sample-path pseudo-  
638 regret 585 and against  $\sqrt{KT} = 632$ .

639 Interpretation. Estimating  $P$  is nearly free on this family, and the gap between the analysis and the  
640 behavior is an order of magnitude, so the analysis rather than the method looks conservative.

641 Limitation. The variant run here has no guarantee of its own, so the comparison is against a rate  
642 proved for a different algorithm. That  $\gamma = 0$  wins is a statement about this family rather than about  
643 the algorithm, which the next study tests.

644 **Where does the plug-in fail?** The family above is mild. Five deliberately hostile maps at  $K = 40$ ,  
645  $|\mathcal{S}| = 8$ ,  $T = 8000$ ,  $d = 50$  and 30 paired seeds, with  $\kappa$  measured on each: a best action reachable  
646 only through a state the others rarely produce, a sensor whose mass sits on one state, a state of  
647 probability  $10^{-3}$  everywhere, a deterministic map, and one whose good action's early visits are  
648 unrepresentative.

| environment             | $\kappa$ | known $P$ | action-level | plug-in at mixing rate $\gamma$ |       |        |        |
|-------------------------|----------|-----------|--------------|---------------------------------|-------|--------|--------|
|                         |          |           |              | 0                               | 0.01  | 0.05   | 0.20   |
| rare optimal action     | 125.9    | 950.7     | 1302.3       | 948.2                           | 976.8 | 1087.8 | 1503.0 |
| unbalanced coverage     | 152.7    | 483.4     | 604.4        | 474.1                           | 476.2 | 487.0  | 531.8  |
| near-unreachable state  | 108.8    | 552.9     | 650.4        | 559.5                           | 563.3 | 575.8  | 624.7  |
| deterministic rows      | 2.5      | 556.3     | 733.7        | 571.9                           | 585.9 | 653.9  | 904.3  |
| misleading early visits | 1.5      | 674.0     | 847.2        | 679.9                           | 688.0 | 716.8  | 834.3  |

650 Observed. The plug-in beat action-level weighting on all five, including two with  $\kappa$  above 100, and  
651 forced exploration cost regret in every cell. At the Laplace default  $\alpha = 1$  the deterministic map is  
652 the one case where the plug-in *loses* to action-level weighting, 750.3 against 733.7. Sweeping  $\alpha$   
653 over 1, 0.3, 0.1, 0.03, 0.01 gives 750.3, 642.5, 601.4, 581.1, 571.9 there and is monotone on all five,  
654 and  $\alpha = 0.01$  is what the tables above use.



655 Interpretation. The mixing rate that Theorem 9 tunes is doing analytic work rather than algorithmic work. The single loss has an identifiable mechanism, since one visit to a deterministic row is  
656 recorded as  $\hat{P}(s \mid a) = 2/(1 + |\mathcal{S}|)$  rather than 1, so rarely played actions are scored as if spread  
657 over states they never reach.  
658

659 Limitation. The leading rate is indifferent to fixed  $\alpha$  and the algorithm is not, which is worth stating  
660 rather than tuning away. These five maps show that hostility to the plug-in is survivable at a  
661 small constant, not that it is harmless, and no map adversarial to the smoothing constant itself was  
662 constructed.

663 **Does safe blending recover the pooling gain, and if not why not?** The rule of Theorem 11, run  
664 in the  $m = 1$  variant like everything else in this appendix rather than in the analyzed  $m = d + 1$   
665 one, was compared against its two ingredients at  $K = 40$ ,  $|\mathcal{S}| = 8$ ,  $d = 50$  and 20 paired seeds.  
666 Measuring the realized bias against the certificate’s own budget locates the outcome. At  $K = 40$ ,  
667  $|\mathcal{S}| = 8$ ,  $T = 8000$ ,  $d = 50$  and 12 seeds the per-round budget  $\frac{\eta}{2}(K - 2|\mathcal{S}|)$  sums to 307.3 over the  
668 horizon, and

|     | summed  | versus budget | quantity                                                                |
|-----|---------|---------------|-------------------------------------------------------------------------|
|     | 128.7   | 0.42×         | what the regret charges, $\sum_t  \langle x_t, b_t \rangle - b_t(a^*) $ |
| 669 | 513.7   | 1.67×         | what Proposition 8 charges, $\sum_t \max_a  b_t(a) $                    |
|     | 32000.0 | 104×          | what the algorithm can verify, Hoeffding radii                          |
|     | 32000.0 | 104×          | the same with Bernstein radii                                           |

670 Observed. The rule reproduced action-level regret to the digit in every cell, 533.2, 754.4 and 1317.0  
671 at  $T = 8000$  on the Dirichlet, deterministic and rare-optimal families and 2304.6, 2023.5 and 3108.8  
672 at  $T = 50000$ , against a plug-in that scores 464.8, 585.5, 956.5 and 1770.6, 1473.5, 2346.6 on the  
673 same runs. The fraction of round-action pairs it cleared for pooling was 0.000 everywhere, at both  
674 horizons, at both smoothing constants, and with the confidence logs dropped as well.

675 Interpretation. The implemented certificate selected no pooling in any run, so the policy coincided  
676 numerically with the action-level baseline rather than merely matching its guarantee, and the table  
677 says where that comes from. On this trajectory the realized bias lies below the best-case variance  
678 budget, so pooling would pay for itself here. The bound of Proposition 8 is 1.67 times the budget  
679 with a factor of 4.0 available between it and the truth, so a tight bias bound would close the analytical  
680 gap. What the algorithm can verify is saturated at the trivial cap of four per round, and Bernstein  
681 radii buy nothing over Hoeffding ones, which is where the failure lives.

682 Limitation. Two exact refinements of the bias bound, an oscillation in place of a maximum and a  
683 Cauchy–Schwarz split into the comparator’s effective dimension times a chi-square, give 577.9 and  
684 5931.6 against a truth of 94.1 over  $T = 4000$ , where the maximum gives 368.0, so both are worse  
685 than what they replace. The budget table is one diagnostic trajectory rather than a worst case, so it  
686 shows this certificate inert here and not that no certificate can clear.

687 **Can the denominator be counted instead of predicted?** With  $x$  frozen over blocks of 400 and the  
688 states of the first half counted, at  $K = 40$ ,  $|\mathcal{S}| = 8$ ,  $T = 8000$ ,  $d = 50$  and 12 seeds, the counted  
689 denominator gives regret 496.5 against 499.5 for the predicted one and a comparator bias of 99.6  
690 against 52.9, both under the budget of 153.7 for the pooled half.

691 Observed. Counting is the noisier of the two on regret and on bias. Computed under uniform play,  
692 the predicted relative error is vacuous at every setting tried,  $K = 40$  through  $K = 4000$ , while the  
693 counted one reads 0.0244 against a budget of 0.0892 at  $K = 1000$  and  $T = 10^5$ .

694 Interpretation. A prediction pools every action’s history while a count uses one block, which is why  
695 counting is noisier, and yet the counted certificate is the better scaled of the two. That setting is the  
696 first in which any certificate for  $\rho_t$  clears at all.

697 Limitation. The window that buys it is 53 per cent of the horizon, and the frozen-block regret at that  
698 window,  $1.2 \times 10^7$ , exceeds the  $2.6 \times 10^4$  the pooling was meant to improve, so what clears is the  
699 certificate and not a usable algorithm.

700 **Does the effective dimension, rather than  $|\mathcal{S}|$ , govern the gain?** Holding  $|\mathcal{S}| = 8$ ,  $K = 40$ ,  
701  $T = 5000$ ,  $d = 20$  and 24 seeds, and varying only the overlap between the rows of  $P$ , the measured  
702 mean of  $v_t$  moves while  $|\mathcal{S}|$  does not:

| row overlap | mean $v_t$ | state-pooled | action-level | paired gain      |
|-------------|------------|--------------|--------------|------------------|
| 0.00        | 2.719      | 392.7        | 599.3        | 206.6 $\pm$ 20.0 |
| 0.25        | 2.013      | 361.1        | 528.1        | 167.0 $\pm$ 21.3 |
| 0.50        | 1.532      | 306.5        | 406.5        | 100.0 $\pm$ 10.5 |
| 0.75        | 1.168      | 199.9        | 229.2        | 29.3 $\pm$ 3.2   |
| 0.95        | 1.009      | 49.2         | 50.8         | 1.6 $\pm$ 0.5    |

Observed. Both the regret and the advantage over action-level weighting fall monotonically with  $v_t$ , from a paired gain of 206.6 at  $v_t = 2.719$  to 1.6 at  $v_t = 1.009$ . Nothing about the raw state count changes across the sweep.

Interpretation.  $|\mathcal{S}|$  is constant and cannot explain the pattern, while  $v_t$  moves with it, which is the comparison Theorem 1 predicts.

Limitation. The two estimators do not become the same object at  $v_t \approx 1$ . Rather the rows of  $P$  agree, so every action induces the same loss and both regrets collapse, making that endpoint uninformative rather than favorable. One parameter of one map family was varied, so the sweep is consistent with the effective dimension governing the gain rather than establishing it.

**Is there a bias-variance trade in the representation?** With 16 raw states drawn from four latent clusters,  $K = 40$ ,  $T = 6000$ ,  $d = 20$  and 24 seeds, running the algorithm on nested groupings of size 1, 2, 4, 8, 16 gives, for the analyzed  $m = d + 1$  variant, regret 1590, 831, 754, 920, 1080 at within-cluster heterogeneity 0, with mean  $v_t$  rising 1.000, 1.122, 1.435, 1.663, 2.089 and  $\delta_t(g)$  falling 0.700, 0.275, 0, 0, 0.

Observed. The minimum sits at four groups, the number of latent clusters, and stays there at heterogeneity 0.12, where regret reads 1657, 885, 787, 960, 1113. Sample-path pseudo-regret stayed below the bound of Theorem 2 in every cell. The bound's own optimum agrees at heterogeneity 0, where it reads 9364, 4321, 1155, 1243, 1394 and also selects four groups, but not at 0.12, where it reads 11128, 6073, 3430, 3231, 1385 and selects the finest representation.

Interpretation. A trade-off in the representation exists and is located at the latent structure, which is what Theorem 2 describes qualitatively.

Limitation. The bias term  $2 \sum_t \delta_t(g)$  is the conservative half of the trade, so the theorem certifies a trade-off whose location it does not predict once groups are heterogeneous, and the latent cluster count was known to the experimenter and not to any algorithm.

Separately, on the construction of Theorem 3 with  $h = d$  the mean pseudo-regret over 400 seeds is 44.1 against a predicted fluctuation of 40.5, and of the five policies compared there the four without access to the block signs attain expected loss within 0.05% of  $T/2$ , as Lemma 14 predicts, while the one given them attains pseudo-regret  $-954$ . The code producing every reported number and figure is at <https://github.com/melikabaghi/state-exp3>. It pins the software environment used for the reported numbers, python 3.14.3 with numpy 2.4.2 and matplotlib 3.10.8, states the seed count for each study, lists one command per numbered study, and commits the stored outputs each figure reads so that every figure rebuilds without rerunning an experiment.

**Does the drift bound describe the right scaling?** The construction of Theorem 3 at  $K = 40$ ,  $T = 8000$ ,  $h = d$  and 12 paired seeds, sweeping  $d$  from 10 to 200, the amplitude  $\varepsilon$  from 0.02 to 0.20 so that  $E_2$  runs from 12 to 1193, and the drifting coordinates  $q$  from 1 to 16. Four policies are run, uniform, action-level EXP3, STATE-EXP3, and the greedy policy on the exact stale vector  $m_t$ , the last because Theorem 3 claims to survive it.  $E_2$  is measured on the realized instance rather than predicted, so the normalization uses the quantity the theorem uses.

Observed. The predicted scale  $\sqrt{d E_2 \min\{1 + \log q, T/d\}}$  ranges from 38 to 377 across the grid and the measured regret tracks it (Figure 3). Normalized, the four policies read 0.230 to 0.429, 0.228 to 0.424, 0.238 to 0.417 and 0.237 to 0.437, a spread below 1.9 against a tenfold range in the predictor. The oracle-assisted policy has the largest mean ratio, 0.345 against 0.311, 0.314 and 0.310 for the three that never see  $m_t$ .

Interpretation. The three arguments enter the predicted scale differently and the collapse holds in all three sweeps, so it is the form of the bound rather than one fitted constant that tracks the data. That greedy play on the exact  $m_t$  sits at the top of the band rather than below it is the concrete content of the claim that the bound survives the oracle.

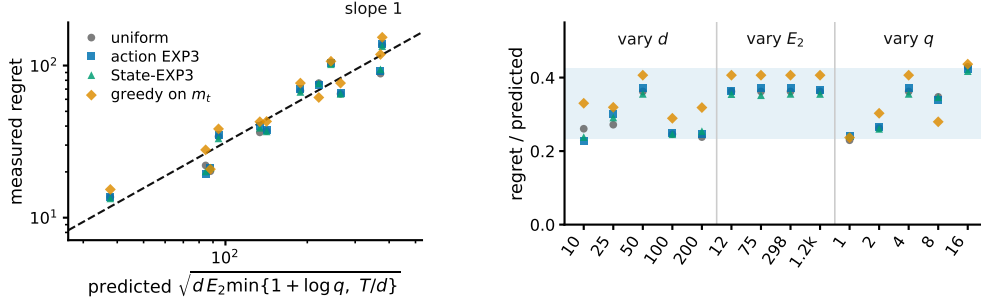


Figure 3: Study 8, the drift channel. **Left**, measured regret against the scale Theorem 3 predicts, over twelve cells in which  $d$  varies 20-fold,  $E_2$  100-fold and  $q$  16-fold. The dashed line has slope one. **Right**, the same points divided by that scale, grouped by which parameter the cell sweeps, with the shaded band spanning the observed range. STATE-EXP3 appears here only as one more algorithm on the hard family, where it has no advantage.

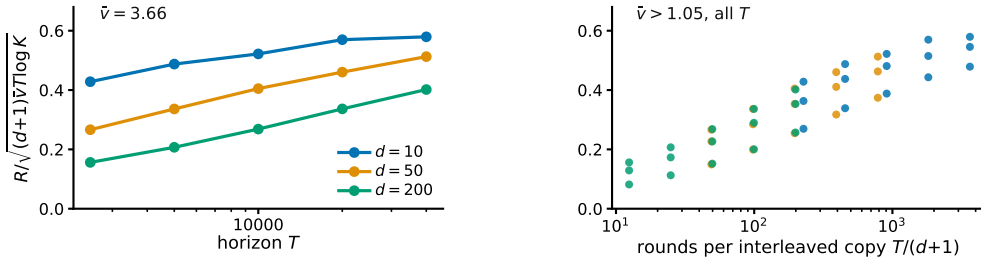


Figure 4: Study 9, the state channel. The plotted ratio would be flat if Theorem 1 captured the dependence on all three of  $T$ ,  $d$  and  $\hat{v}$ . **Left**, horizon sweep at the largest effective dimension, one line per delay. **Right**, every cell with  $\hat{v} > 1.05$  against rounds per interleaved copy, where the three delays fall on one curve.

751 Limitation. Every cell has  $T/d > 1 + \log q$ , so only the  $1 + \log q$  branch of the minimum is  
752 exercised and the  $T/d$  saturation branch, which needs  $d$  comparable to  $T$ , is not probed. A lower  
753 bound quantifies over all algorithms and four policies cannot establish that. What these runs check  
754 is the scaling of the construction.

755 **Does Theorem 1 describe the right dependence on  $T$ ,  $d$  and  $\hat{v}$ ?** A  $4 \times 3 \times 5$  grid in row overlap,  
756 delay and horizon at  $K = 40$ ,  $|\mathcal{S}| = 8$  and five paired seeds, running the analyzed  $m = d + 1$  variant  
757 and reporting  $R/\sqrt{(d+1)\hat{v}T \log K}$ . A raw regret table cannot answer this and neither can a bound  
758 that holds with room to spare, since the question is about dependence rather than level.

759 Observed. Over the full grid the ratio spans 35.9 times, from 0.016 to 0.579. Dropping the near-  
760 degenerate column at  $\hat{v} = 1.013$  leaves 7.1 times over 45 cells. At  $\hat{v} = 3.656$  and  $d = 10$  it moves  
761 only 1.35 times across a sixteenfold range of  $T$ , and at  $d = 200$  it moves 2.57 times. Against rounds  
762 per interleaved copy  $T/(d+1)$  the three delays fall on one increasing curve (Figure 4).

763 Interpretation. The data are consistent with the predicted  $\sqrt{T}$  dependence once each interleaved  
764 copy receives enough rounds, while much of the remaining variation is associated with the number  
765 of rounds available to each copy. At  $d = 200$  and  $T = 2500$  each of the 201 copies sees 12 rounds,  
766 so that cell is nowhere near the regime the bound describes. Together with the  $m = 1$  comparison in  
767 Study 1, this suggests that much of the observed delay penalty comes from the interleaving device  
768 used in the analysis rather than from the pooled estimator itself.

769 Limitation. The degenerate column is degenerate for the reason study 6 gives, since at  $\hat{v} \approx 1$  the  
770 rows of  $P$  agree, every action induces nearly the same loss, and both regrets collapse for a reason  
771 the bound cannot see. One natural explanation for the remaining looseness is ruled out rather than

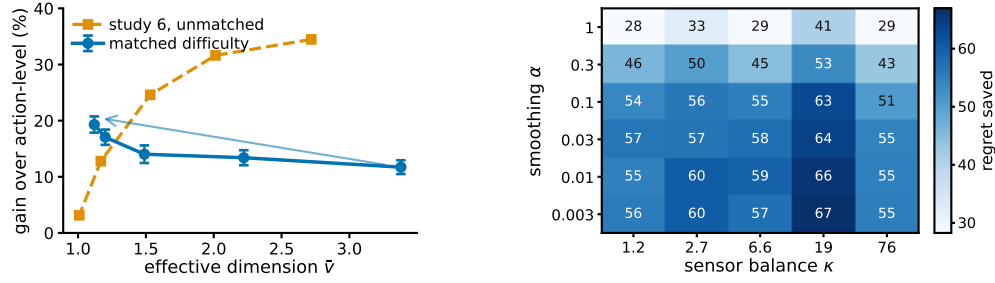


Figure 5: **Left**, study 10. The same overlap sweep with and without holding the task difficulty fixed. Matching the uniform-play regret reverses the trend, supporting the interpretation that the decline in Study 6 is driven by task-difficulty confounding rather than by the pooling mechanism itself. Bars are one standard error of the paired difference. **Right**, study 11, regret saved against action-level weighting over the smoothing pseudocount and the sensor balance, 10 paired seeds per cell.

confirmed, since the realized  $\sum_t v_t$  lies within 1.01 to 1.49 of the budget  $\hat{v}T$ , so the sup over play distributions is not the source.

**Does pooling help because the rows overlap, or because overlap made the task easy?** Study 6 sweeps the overlap and finds the advantage falling as  $\hat{v}$  falls, which read at face value contradicts Theorem 1, whose bound improves as  $\bar{v}$  falls. That sweep is confounded, since raising the overlap also flattens  $c_t = P\theta_t$  across actions, collapsing the comparator gap so that both regrets vanish together. This study holds  $K = 40$ ,  $|\mathcal{S}| = 8$ ,  $T = 10000$ ,  $d = 50$  and the difficulty fixed, and varies only  $\hat{v}$ . Difficulty is the uniform-play regret  $\sum_t \text{mean}_a c_t(a) - \min_a \sum_t c_t(a)$ , a property of the instance with no algorithm in it, and the amplitude of  $\theta$  is bisected in every cell to hit a common target. The map keeps one contrast direction alive at every overlap and  $\theta$  is aligned with it, which is what lets the gap survive as the rows agree.

Observed. Across 20 paired seeds the relative gain rises from 11.7 per cent at  $\hat{v} = 3.380$  to 19.3 per cent at  $\hat{v} = 1.120$ , with absolute gains  $85.5 \pm 8.9$  to  $106.2 \pm 7.9$  and a correlation of  $-0.929$  between  $\hat{v}$  and the gain. Study 6’s unmatched sweep falls from 34.5 to 3.1 per cent over a comparable range (Figure 5).

Interpretation. The two sweeps disagree in sign, and the matched-difficulty sweep agrees with the overlap dependence suggested by Theorem 1, although this experiment uses the practical  $m = 1$  variant rather than the theorem’s  $m = d + 1$  algorithm. Overlap is what pooling exploits, and study 6 measures overlap confounded with an instance becoming trivial. This also explains study 9’s degenerate column without appealing to it.

Limitation. At the two highest overlaps some seeds cannot reach the common target and the realized difficulty is 5 to 13 per cent below it, so the matching is close rather than exact. Since a smaller gap should if anything shrink the advantage, the measured rise is conservative in that direction. The reversal is one construction at one target, and the matched gains are also smaller in magnitude than the unmatched ones, which is what fixing the difficulty costs.

**Is the plug-in fragile in the smoothing constant, and does that depend on coverage?** Study 3 reports one map where the plug-in loses at  $\alpha = 1$  and recovers as  $\alpha$  falls, which is an anecdote about a single instance. Here  $\alpha$  is swept over 1 to 0.003 against the sensor balance  $\kappa$  of Theorem 9, controlled from 1.2 to 76 by placing mass  $\rho$  on one scarce state and drawing rows around that mean, at  $K = 40$ ,  $|\mathcal{S}| = 8$ ,  $T = 8000$ ,  $d = 50$  and 10 paired seeds.

Observed. No cell loses to action-level weighting, the gain running from 28.3 to 66.9 over the 30 cells. The gain roughly doubles as  $\alpha$  falls from 1 to 0.03 and is flat below that, while moving along  $\kappa$  at fixed  $\alpha$  changes it far less, the worst-to-best plug-in regret within a  $\kappa$  column being 1.07. At the better  $\alpha$  the plug-in is indistinguishable from known  $P$ , the paired difference clearing two standard errors in one of the five columns and changing sign in another.

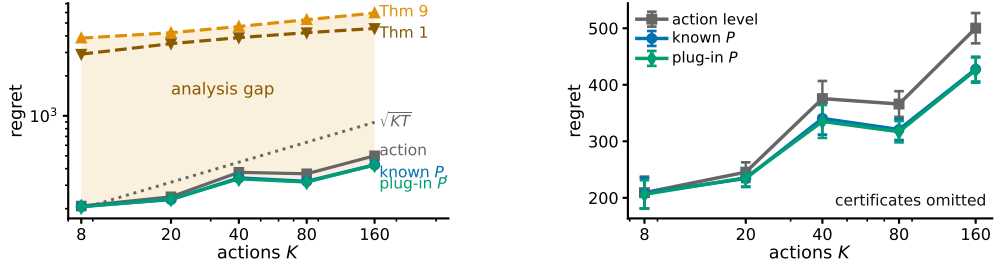


Figure 6: Study 12. **Left**, the three measured arms and the two certificates on one axis,  $|\mathcal{S}| = 8$ ,  $T = 5000$ ,  $d = 50$ , 12 paired seeds, every measured arm at  $m = 1$ . The shaded region is what the analysis costs rather than what the method does. **Right**, the same measured arms rescaled with the certificates removed, which is the only way to see that known  $P$  and the plug-in coincide. Bars are one standard error.

807 Interpretation. Of the two parameters the analysis names, the one that matters here is the one the  
 808 analysis treats as free. A Laplace default costs about half the available gain at every coverage  
 809 level tried, and coverage imbalance over a sixtyfold range costs much less than that. Taken with  
 810 study 2 this is further evidence that the difficulty in the unknown- $P$  setting is certification rather  
 811 than estimation.

812 Limitation. This is one map family with  $\kappa$  varied through a single scarce state, so it does not rule  
 813 out a construction in which  $\kappa$  is the binding parameter, and Theorem 9 still carries  $\kappa$  for a worst case  
 814 this family need not contain. The flatness below  $\alpha = 0.03$  says the leading behavior stops moving,  
 815 not that any fixed  $\alpha$  is safe at every horizon.

816 **How far is the method from its certificate, and the certificate from useful?** Study 2 reports these  
 817 numbers in separate columns, leaving the reader to assemble them. Here action-level weighting,  
 818 STATE-EXP3 with  $P$  given, and the same estimator with  $\hat{P}_t$  updated every round are measured on  
 819 the same paired seeds across  $K$  from 8 to 160, and drawn beside the two certificates. Theorem 1 is  
 820 the certificate for known  $P$  and analyzes  $m = d + 1$ , while Theorem 9 is the warm-start rate and  
 821 analyzes an algorithm that freezes  $\hat{P}$  and restarts, so both are plotted to show the size of a gap and  
 822 not as like for like.

823 Observed. The plug-in stays within 1.6 per cent of known  $P$  at every  $K$ , the two curves being  
 824 visually indistinguishable, while action-level weighting costs 1.4 to 17.4 per cent more than the  
 825 plug-in as  $K$  rises twentyfold. Theorem 1 sits 10.7 to 14.8 times above the measured plug-in and  
 826 Theorem 9 sits 14.0 to 18.7 times above it. The warm-start certificate also stays above  $\sqrt{KT}$  at  
 827 every  $K$  tried, by factors of 19.2 down to 6.6.

828 Interpretation. The vertical gap reflects analytical conservatism rather than observed algorithmic  
 829 performance. Estimating  $P$  online is close to free on this family, so the order of magnitude between  
 830 the measured arms and either certificate is conservatism in the proof. That the warm-start rate never  
 831 drops below  $\sqrt{KT}$  here is the concrete form of the statement in Section 4 that it does not certify the  
 832 channel it analyzes.

833 Limitation. The measured arms run  $m = 1$  and Theorem 1 analyzes  $m = d + 1$ , and the plug-in  
 834 is not the warm-start algorithm Theorem 9 analyzes, so neither vertical gap is a statement about a  
 835 bound being loose for the algorithm it covers. A conservative bound and a well-behaved algorithm  
 836 are consistent with a third possibility, that the family is easy.

837 **Can the coarsening be chosen from data?** Theorem 2 takes the representation as given and study 7  
 838 uses the latent cluster count, which no algorithm knows. This study supplies a rule and measures  
 839 what it costs, with no theorem behind it. A prefix of 1200 rounds runs the identity representation  
 840 and estimates  $\hat{\theta}(s)$  by averaging the outcomes seen at each state. Sorting states by  $\hat{\theta}$  and merging  
 841 adjacent pairs greedily gives one candidate at every granularity from 16 down to 1, each scored  
 842 by the quantity Theorem 2 bounds,  $\sqrt{2(d+1)V(g)\log K} + 2\sum_t \hat{\delta}_t(g)$ , and the argmin runs the

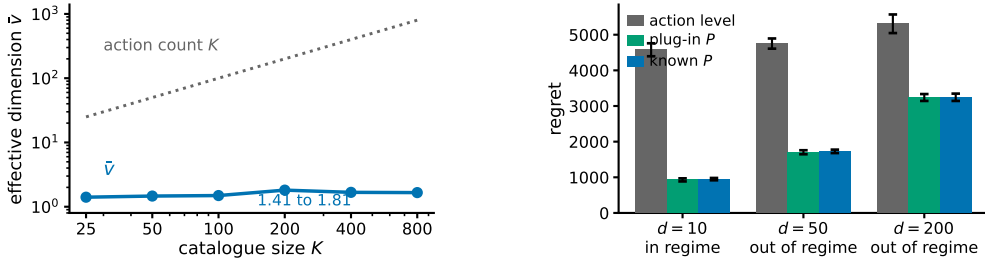


Figure 7: Study 14, a recommendation funnel. **Left**, the effective dimension stays near two while the catalogue grows thirty-two-fold, so the gap between  $K$  and  $\hat{v}$  comes from the catalogue having categories, not from a chosen  $K$ . **Right**, the three arms at  $T = 20000$  and 8 paired seeds, labelled by whether the regime condition of Section 4 holds. Bars are one standard error. No real data is used anywhere in this study.

843 remaining rounds. Against it are the identity, the true clusters, and the best candidate on the ladder  
844 scored in hindsight, over 24 paired seeds.

845 Observed. The rule fails. It selects 15 groups at heterogeneity 0 and 16 at 0.12, against a true cluster  
846 count of four, and lands at  $912.9 \pm 30.1$  and  $944.7 \pm 31.1$  where the identity gives  $921.0 \pm 30.5$  and  
847  $952.7 \pm 32.4$ , so it captures under one per cent of the available gain. The true clusters give  $843.7 \pm$   
848  $27.9$  and  $883.0 \pm 31.1$ , and hindsight over the same ladder gives  $833.5 \pm 27.8$  and  $841.5 \pm 23.6$ .

849 Interpretation. Hindsight nearly matches the true clusters, so the ladder does contain a good repre-  
850 sentation and the failure is the criterion rather than the candidate set. Measuring  $\hat{\delta}$  at the four-group  
851 representation locates it. At heterogeneity 0 the true within-group spread is exactly zero while the  
852 prefix estimates it at 0.154, which is sampling noise, and the bound multiplies that by  $2(T - T_0)$  to  
853 give a phantom bias of 1483 against a real variance saving of 1261. The bias term is the conservative  
854 half of Theorem 2, and a noisy plug-in estimate of it is enough to forbid every merge.

855 Limitation. One selector on one family, and a negative result about this rule is not a negative result  
856 about data-driven selection. The obvious repairs, debiasing  $\hat{\delta}$  for its sampling noise or scoring  
857 candidates on held-out loss instead of on the bound, are not evaluated here. The prefix is also charged  
858 to every arm equally, so the comparison understates what a selector that spent less on estimation  
859 might do.

860 **Is “many actions resolved by few states” a natural regime?** Every study so far draws  $P$  from  
861 a Dirichlet, which is the assumption under test rather than evidence for it. This one builds an  
862 environment to the shape of a recommendation funnel instead. Actions are  $K = 200$  catalogue  
863 items, states are  $|\mathcal{S}| = 6$  ordered engagement depths, and the outcome is a delayed conversion.  
864 Three structural facts are imposed because funnels have them, namely that items fall into a modest  
865 number of categories sharing an engagement profile, that item quality is heavy-tailed, and that the  
866 loss falls with engagement depth and drifts seasonally rather than adversarially. The generated  
867 catalogue puts 74 per cent of its mass on no engagement and 1.4 per cent on the deepest.

868 Observed. The effective dimension is  $\hat{v} = 1.97$  against  $K = 200$ , a hundredfold gap, and it stays  
869 between 1.41 and 1.81 as the catalogue grows from 25 to 800 items (Figure 7). Against action-level  
870 weighting the state-pooled variant saves 79.3, 63.6 and 38.9 per cent at  $d = 10, 50$  and 200, the  
871 largest margins anywhere in this appendix. The plug-in matches known  $P$  at every delay, the paired  
872 difference never reaching two standard errors. The regime condition  $\hat{v}(d + 1) \log K \leq K + d$  holds  
873 at  $d = 10$  and fails at  $d = 50$  and  $d = 200$ .

874 Interpretation. The separation between  $K$  and  $\hat{v}$  arrives from catalogue structure rather than from  
875 a tuned parameter, and it does not close as the catalogue grows, which is the claim the Dirichlet  
876 families cannot make. That the condition Theorem 1 needs fails at the delays a funnel has, while  
877 the practical variant still saves most of the regret, is the same message as study 1 in the setting that  
878 motivates the paper. It is evidence for Conjecture 4 rather than for the theorem.

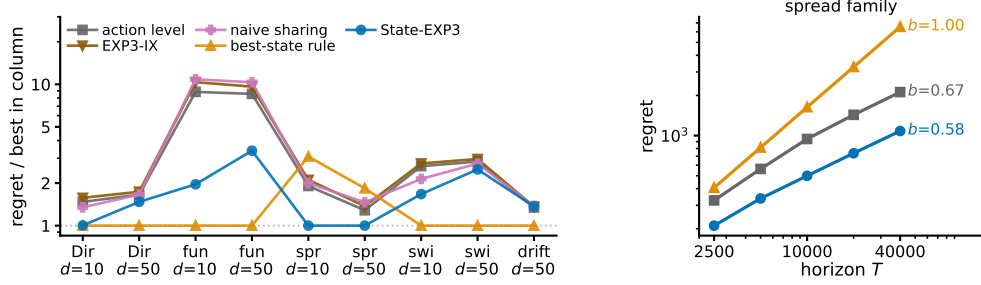


Figure 8: Study 15. **Left**, regret relative to the best arm in each family, so 1 is that column’s winner, over five policies, nine settings and 10 paired seeds. The best-state rule leads seven columns and STATE-EXP3 leads two. **Right**, growth in  $T$  on the one family built so the rule’s assumption is false, with fitted exponents. The left panel is a worst case over families chosen here; the right panel is why that is not a worst case over the model.

879 Limitation. *No real data is used.* There is no interaction log in this environment, nothing here  
880 is fitted to one, and this is not a semi-synthetic benchmark in the usual sense of that phrase. The  
881 structure is imposed by hand from qualitative properties, so it shows that the regime is describable  
882 and self-consistent, not that any deployed system occupies it. The arms run  $m = 1$ , which carries no  
883 guarantee, and the environment has no adversarial component, so it exercises the pooling channel  
884 and not the drift channel.

885 **Which assumption produces which benefit?** Comparing against action-level weighting alone  
886 leaves open whether the gain comes from the state structure or from an implementation detail. Five  
887 policies are run on the same paired seeds over nine settings. Besides action-level weighting and  
888 STATE-EXP3 they are EXP3-IX (Neu, 2015), the implicit-exploration variant of EXP3, an im-  
889 portance weight; *naive sharing*, which credits  $P(S_t | a)X_t/\bar{p}(S_t)$  using the unconditional state  
890 distribution and so isolates the value of the denominator  $q_t = x_t^\top P$  from the value of sharing at all;  
891 and a *best-state rule*, which estimates  $\hat{\theta}(s)$  online, takes the best state, and plays the action most  
892 likely to reach it. Two settings are added to the earlier families, one where the best action mixes  
893 over several good states rather than maximizing the single best one, and one that is obliviously non-  
894 stationary. Running a bandit over states and mapping the values back through  $P$  is not a separate  
895 baseline, since  $\sum_s P(s | a)1\{S_t = s\}X_t/q_t(s)$  equals  $P(S_t | a)X_t/q_t(S_t)$  identically, so that  
896 method is STATE-EXP3.

897 Observed. STATE-EXP3 does not win most columns. The best-state rule leads seven of the nine  
898 and STATE-EXP3 leads two, and over these nine settings the rule is also better on the median, 1.00  
899 against 1.47 times the column winner, and on the worst case, 3.08 against 3.39 (Figure 8). EXP3-IX  
900 is 1.5 to 17.1 per cent worse than plain action-level weighting everywhere, and naive sharing ranges  
901 from 18.7 per cent better to 22.7 per cent worse than it. On the spread family the ordering inverts,  
902 STATE-EXP3 leading by 47.6 and 22.2 per cent while the best-state rule falls 61.5 and 43.0 per  
903 cent behind action-level weighting. Sweeping the horizon there, the rule’s regret grows with fitted  
904 exponent 1.00 and STATE-EXP3’s with 0.58.

905 Interpretation. The best-state rule buys its lead with an assumption, that the best action is the one  
906 most likely to reach the single best state. Where that holds it is very strong, and in the funnel it holds  
907 in 10 of 10 seeds because engagement depth is ordered, which is why it leads there by a factor of two.  
908 Where it fails the rule commits to a fixed action and pays regret linear in  $T$ , which no exploration  
909 rate repairs, and the exponent of 1.00 is that happening. The comparison therefore separates the  
910 assumptions rather than ranking the methods. STATE-EXP3 assumes nothing about the shape of  
911  $\theta$ . Across these nine selected settings its largest regret ratio to the column winner is 3.39, and the  
912 rule’s suite maximum is slightly smaller at 3.08, but the rule’s fitted growth on the spread family is  
913 linear. Action-level weighting also carries a sublinear guarantee, so what separates the arms here is  
914 the constant rather than the rate. STATE-EXP3’s paired standard errors are the smallest of the five at  
915 5.8 per cent of the mean against 23.6 for the rule. That naive sharing can be worse than no sharing  
916 is the separate point that the denominator, not the sharing, is what Proposition 6 buys.



Limitation. Nine settings chosen here are not a worst case over the model, and the median and worst-case columns above should be read as describing this set only. A practitioner whose problem has the ordered structure of the funnel should use the best-state rule, and this appendix does not identify when that structure can be verified in advance. On the drift family no arm's paired difference from action-level weighting reaches two standard errors, the largest being the best-state rule's  $+24.6 \pm 12.4$ , so that column is reported as an absence of signal rather than a result.

Taken together the fifteen studies cover both channels. Studies one, six, seven, nine, ten, fourteen and fifteen probe the state channel. They are consistent with the effective dimension, rather than  $|\mathcal{S}|$  or  $K$ , governing what pooling buys, with the qualification that study ten had to control for task difficulty before study six's sweep pointed the way Theorem 1 does, and study nine shows where the theorem stops describing the dependence and attributes that to the interleaving. Studies two to five, eleven and twelve probe what happens when the action-to-state matrix must be estimated, and locate the difficulty in certifying the estimate rather than in forming it, study eleven adding that the smoothing constant matters more than the coverage imbalance the analysis carries and study twelve putting the measured arms and their certificates on one axis. Study thirteen closes the representation thread by showing that the bound of Theorem 2 is too conservative to serve as its own selection rule. Study fourteen is the only one whose action-to-state matrix is not drawn from a Dirichlet, and it reports the largest margins here, though on a structure imposed by hand rather than measured. Study fifteen widens the baseline set and finds STATE-EXP3 beaten in seven of nine settings by a heuristic that assumes the best action is the one reaching the single best state, which holds on several of these families and fails with linear regret when it does not. Study eight is the drift channel, and its collapse across  $d$ ,  $E_2$  and  $q$  is the strongest evidence here that Theorem 3 has the right form. No study runs a method that uses both channels, which is the gap Section 8 describes.

## J Why the delay does not become additive

Run STATE-EXP3 with  $m = 1$ , so  $x_t \propto \exp(-\eta L_t)$  with  $L_t = \sum_{r \leq t-d-1} \hat{c}_r$ . Let  $\tilde{x}_r$  be the iterate that has also absorbed the outstanding estimates  $G_r = \sum_{s=r-d}^{r-1} \hat{c}_s \geq 0$ . Two terms appear.

The delay term is  $\mathbb{E}[\langle x_r - \tilde{x}_r, \hat{c}_r \rangle]$ . Since  $\tilde{x}_r(a) \geq x_r(a)e^{-\eta G_r(a)}$  and  $1 - e^{-z} \leq z$ , it is at most  $\eta \mathbb{E}[\sum_a x_r(a) G_r(a) \hat{c}_r(a)] = \eta \sum_{s=r-d}^{r-1} \mathbb{E}[\langle x_r, \hat{c}_s \rangle]$ , and  $x_r$  depends only on rounds  $r-d-1$  and earlier, hence on rounds before  $s$ , so  $\mathbb{E}[\langle x_r, \hat{c}_s \rangle] = \mathbb{E}[\langle x_r, c_s \rangle] \leq 1$ . This term is  $\eta dT$  for any unbiased estimator, which is the additive form.

The quadratic term is where it stops. The telescoping scores  $\sum_a x_{r+d}(a) \hat{c}_r(a)^2$ , with  $x_{r+d}$  the iterate that has absorbed rounds up to  $r-1$ , while Proposition 6 bounds the play-weighted sum, the one against  $x_r$ , by  $v_r$ . Carrying out the substitution,  $\mathbb{E}[\sum_a x_{r+d}(a) \hat{c}_r(a)^2 \mid \cdot] \leq \sum_s q_{r+d}(s)/q_r(s) \leq |\mathcal{S}|/Z_r$  for  $Z_r = \mathbb{E}_{a \sim x_r}[e^{-\eta G_r(a)}]$ , and  $Z_r \geq e^{-\eta \langle x_r, G_r \rangle}$ , so the swap costs an exponential moment of  $\langle x_r, G_r \rangle$ .

No bound on that moment holds uniformly over state distributions. For this estimator  $\langle x_r, \hat{c}_s \rangle = q_r(S_s)X_s/q_s(S_s)$ , so the quantity is governed by  $1/q_s(S_s)$ , the reciprocal probability of the state actually observed. Its mean is  $\sum_s q_s(s)/q_s(s) = |\mathcal{S}|$ , which is why the delay term is fine. For a fixed  $q$  of full support the exponential moment is finite,  $1/q(S)$  being bounded, and what fails is uniformity. Already at  $|\mathcal{S}| = 2$  with  $q = (\varepsilon, 1 - \varepsilon)$ ,  $\mathbb{E}[e^{\eta/q(S)}] \geq \varepsilon e^{\eta/\varepsilon} \rightarrow \infty$  as  $\varepsilon \rightarrow 0$  while the mean stays at 2, so no bound in  $|\mathcal{S}|$  and  $\eta$  alone is available, and  $q_s$  is set by the algorithm's own past play. Measured at  $|\mathcal{S}| = 4$ , the mean stays at 4.0 while  $\mathbb{E}[e^{0.05/q(S)}]$  runs 1.22, 1.43 and then  $1.3 \times 10^{21}$  as the smallest state probability falls through 0.25, 0.019 and 0.0009.

Flooring  $q$  restores the moment and costs the product back. Mixing with the uniform action at rate  $\gamma$  gives  $q_t(s) \geq \gamma \bar{p}(s)$ , so  $\eta \langle x_r, G_r \rangle \leq 1$  needs  $\gamma \asymp \eta d \kappa |\mathcal{S}|$  under the balanced-sensor condition  $\bar{p}(s) \geq 1/(\kappa |\mathcal{S}|)$ , and the mixing cost  $\gamma T$  then reproduces  $\tilde{O}(\sqrt{d \kappa |\mathcal{S}| T \log K})$ . So the obstruction is not an artifact of one device. Empirically it does not bite,  $1/Z_r$  staying below 1.33 at the tuned  $\eta$  over  $d \in \{10, 50, 200\}$ , which is why Conjecture 4 is stated rather than abandoned.



## 965 **K Stationary sensors**

966 *Observation 17.* The multiplicative term  $\sqrt{\min\{|\mathcal{S}|, d\}T}$  is not forced by the known construction,  
 967 and is absent at one endpoint of the present model class.

968 The  $\Omega(\sqrt{\min\{|\mathcal{S}|, d\}T})$  construction of Esposito et al. (2023, Thm. 5.2) re-splits states across ac-  
 969 tions adversarially per block, so its action-to-state map is time-varying and lies outside the model  
 970 class of Section 3. At  $E_2 = 0$  with stationary  $P$  the problem is a stochastic delayed bandit, where  
 971 delay enters additively (Joulani et al., 2013). One route to it is therefore unavailable and the term is  
 972 absent at one endpoint, but neither a matching upper bound for all  $E_2 > 0$  nor a lower bound ruling  
 973 it out is established here.